Department of Computer Science & Information Technology

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Food Donation App

Submitted By

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Food For All/ Donor App

Project Code:

CMPC-401

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Submission Date:

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Project Manager's Signature

Abstract

Food is one of the basic necessities of humans, and it stands first among all basic need's food, shelter, and clothing. It is important as it nourishes the human body sustaining the very existences of humans. However, with the rising population and development of this country, food wastage has risen to a new high. There are many people who wish to donate food to the needy but are unaware of how exactly they can execute that. Our application (Food for All) revolves around helping the needy by connecting NGOs and Donor. The donors shall be able to see a plurality of options by which they can donate. The NGOs will get the details of the persons wishing to donate via our application and thus a network is established between donors, people who aid the donors in donating (NGOs) and the actual needy people to whom the donated item is sent. Our application aims to bring about transparency, clarity and swiftness in the process of donation thus aiming to mitigate prevailing issues in whatever zone it is possible for us to do so.

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Chapter 1: Introduction

1.1 Introduction

This earth is our home. Humans are our family members. We cannot let anyone *sleep hungry*. Let's change the world! **Food for All** is an android application which is a platform for the people to donate daily routine extra food which they wasted in no manners. Our aim is to reduce food waste and feed the hungry people. People can use this application to donate their foods or to collect food for hungry people. Millions of lives are starving for food around us, have a Tap to feed life. Let's work together to bridge the gap and reduce hunger and food waste in our beloved country. The sharp increase in the amount of wastage in terms of food makes the need for charity in terms of donation. A new internet-based application that provides a platform for donating left over food to all needy people/organizations. It provides information about the motivation to come up with such an application, The product is shown to be an effective means of donating food, items and necessary things to the needy persons and also to donor organizations by using this application. Secondly, in restaurants and banquet halls, a lot of food is wasted on daily basis. So, if a restaurant or banquet hall wants to donate food or anything, he generate a request through this application, our rider will pick up the food or whatever donor wants to donate. It shows the potential for avoiding the wastage of food.

Food is the basic need of every living being. Most people don't realize how much food they throw away every day from uneaten leftovers to spoiled produce. Thereby, an android application has been developed through which people can donate items as per their capacity and the application also allows organizations to put up their requests. The majority of the population today uses smart phones with active internet connection, which is the basic requirement for this product to function properly.

1.2 Project Goals

- Reduction of Food Wastage
- Change cultural of the bagging system
- Our major goal is to provide a platform for food donating and people can easily donate their foods, deliver this food to different NGO's and to the needy/hungry people with the help of volunteers.
- Food that would otherwise be thrown out is donated, undisturbed, and safe to the community.
- This application provides a source of communication between different services that are working on the same mission.
- It will increase efficiency and will create awareness among people for using technology in the best way as it should be used.

1.3 Project/Product Feasibility Report

There are many types of feasibilities:

- Technical
- Operational
- Economic
- Schedule
- Specification
- Information
- Motivational
- Legal and Ethical

1.3.1. Technical Feasibility

Food for All is completely an Android based project. Each technology is freely available and technical skills are required and manageable. Time limitation of the project development and the ease of implementation using these technologies are synchronized. From these it is clear that the project Ride Sharing is technically feasible. Main tools and technologies are:

Technologies:

- Java programming
- Firebase/SQLite
- XML

Tools:

- Android Studio

Each technology is freely available and technical skills are required and manageable. Time limitation of the project development and the ease of implementation using these technologies are synchronized. From these it is clear that the project Ride Sharing is technically feasible.

1.3.2. Operational Feasibility

We are sure that the project will be operationally successful and raise no issues regarding different operations it performs. The user can easily operate the system as it will provide a user-friendly environment. Therefore, there will not be any operational problem for the users of the system while using the website. From these it is clear that the project Ride Sharing is Operationally feasible.

This project is much more operational in Pakistan because in Pakistan there is no system which manage the Food donation. The system is as convenient that the non-technical user can use this system & all operation can perform easily from the front end of the system. NGO wants to manage their system online and keep donor updated about their donation and keep their record safe.

1.3.3. Economic Feasibility

The System will follow freeware software standards therefore no cost will be charged from the potential customers. Bug fixing and maintaining task will have an associated cost. From these it is clear that the project Food for All is Economically feasible. The system being developed is economic with respect to user’s point of view. It is cost effective in the sense that has eliminated the paperwork.

1.3.4. Schedule Feasibility

Project developing period is 20 weeks that are enough to complete the project. From these it is clear that the project Ride Sharing is Schedule feasible. Scheduling and planning are the most important thing in project. It plays a vital role in completing the project on time. We divide our project into tasks with proper dependencies and durations and make critical path. We assign the tasks to the members according to their skills.

1.3.5. Specification Feasibility

Specification feasibility analysis has provided us with the study that whether the requirements of the project are unambiguous or not. Every check regarding the project has also been studied in this phase. The limitations of the scope are analyzed in this phase. The hardware and software specifications for the completion of this project are also feasible. We also have studied that all the requirements are complete, verified and valid. From these it is clear that the project Ride Sharing is Specification feasible. We have proper requirements specification to make our project useful and according to user requirements. We have proper requirements specification to make our project that fulfills the users need and easy to use for general public.

1.3.6. Information Feasibility

The feasibility of information must be assessed regarding its completion, reliability, and meaningfulness. The information provided by Food for All to the users are correct because it publicly generates request when someone creates it. From these it is clear that the project Food for All is Informationally feasible. The information that we have gathered is meaningful, reliable and measurable. It can be authenticated once the system is complete.

1.3.7. Motivational Feasibility

Evaluation of the client staff regarding the motivation to perform the necessary steps correctly and promptly must occur. Our potential users are willing to perform the actions necessary to verify and validate the system and its usability.

Donating manually is a huge waste of time and cumbersome task. In the current context it needs a platform that provide platform for donating items on different locations. Food for All is designed to help donors to give destinations easily and in decided time.

NGOs has no systems to manage their cases online all record in traditional manner (paper work) which causes headache to handle lots of files and keep record safe. It is necessary to develop a system which manage the donation online and get rid of paper work and keep users updated their donation. So, we decided to develop the **Donation App** to manage all the donations online.

1.3.8. Legal & Ethical Feasibility

Food for All is freely available and provide the system an open source system. It has only maintenance cost. It will have great impact on peoples. It will guide the peoples about their

donations. This app is freely accessible by any user. From these it is clear that the project Food for All is Legal & Ethically feasible.
Legally and ethically we have rights to use the tools that are used to develop our project
We use tools like Android Studio etc. which are legal in Pakistan.

1.4 Project/Product Scope

The project "Food 4 All" is used to manage wastage of food in a useful way. We have to reduce that food wastage problem through online. If anyone has excessive food, they enter the food quantity details and the address location in that application. The admin maintained the details of food donator. The donator can create the account and whenever they have excessive food, they can login and give request to the NGO's. After the NGO's view the donator request and give the alert message like time to come and collect the food. The NGO's rider collects foods from donator through their rider. After receiving the food from the rider by NGO's the application send message and notification to that donator, admin and rider. If the NGO's not responds to the donator notification will send to the NGO after sometime if the NGO not responding then request sends to admin, then admin sends the request to other NGO's then NGO voluntary accepts the request and send the rider to donor. Donator can also send money to NGO's. Firstly, rider done face authentication then donor give the parcel to donor and Guest request to NGO for food and the NGO rider take the food to Guest. This mobile application can be implement easily by any donor agency in emergency or any disaster situation.

1.5 Tools and Technology with Reasoning

Back-end

- Java programming

We are constructing our application on android with java.

Database

- Firebase/SQLite

Front-end

- XML

Both are used for user interfaces (UIs)

Tools

- Android studio for development

Documentation:

- Microsoft Word
- Draw.io for UML Diagrams

Device: Android Smart Phone with minimum 5.0 lollypop version above.

- Minimum RAM 8 GB for Android studio and SDK installation
- Processor core i7 5th generation
- Internet Connection

1.6 Vision Document

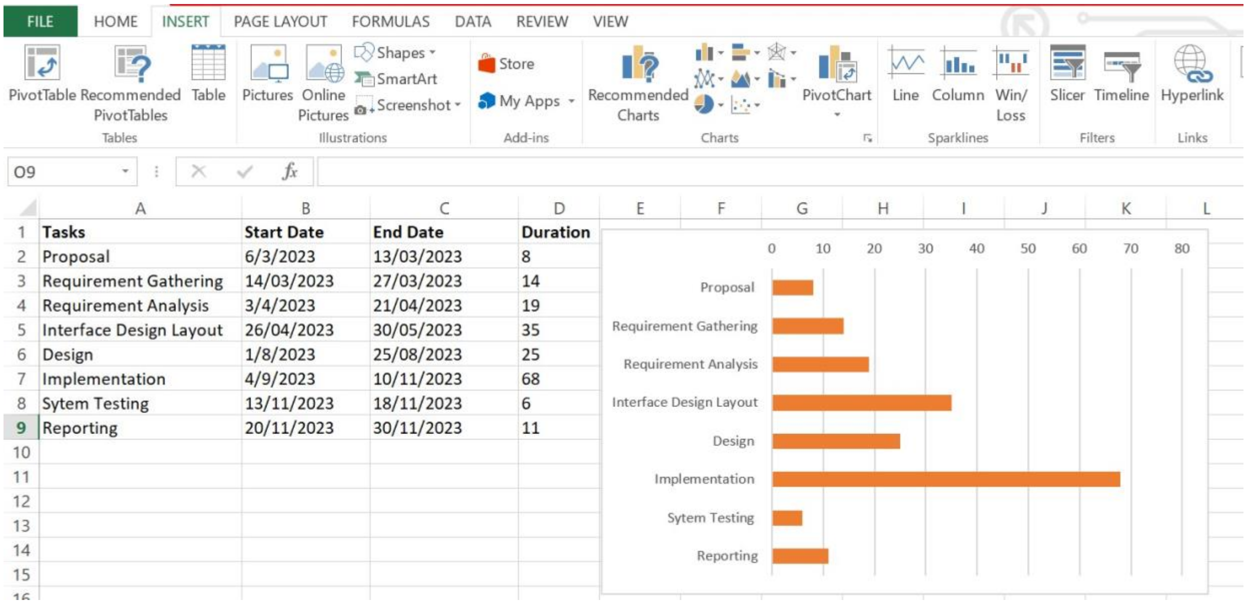
Our aim is to reduce food waste and feed the hungry people. People can use this application to donate their foods or to collect food for hungry people. Millions of lives are starving for food around us, have a Tap to feed life. Let’s work together to bridge the gap and reduce hunger and food waste in our beloved country. The sharp increase in the amount of wastage in terms of food makes the need for charity in terms of donation. A new internet-based application that provides a platform for donating left over food to all needy people/organizations.

Food is the basic need of every living being. Most people don't realize how much food they throw away every day from uneaten leftovers to spoiled produce. Thereby, an Android application has been developed through which people can donate items as per their capacity and the application also allows organizations to put up their requests. The majority of the population today uses smart phones with active internet connection, which is the basic requirement for this product to function properly.

1.7 Product Features/ Product Decomposition

- Benefits will be both the restaurant (reducing food wastage), and the needy
- Keep track of wastage food for restaurant
- User can play role in saving food wastage and help the needy
- Our system is decomposed into two main parts. First one is Android application which is for users such as a donor, rider and NGO. It provides the users with all the functionality such as viewing map, request generation, order acception registration of rider NGO and admin etc. Second is the database which stores information for the users.
- It is our non-profit initiative to serve our nation.

1.8 Gantt chart



Tasks	Start Date	End Date	Duration
Proposal	6/3/2023	13/03/2023	8
Requirement Gathering	14/03/2023	27/03/2023	14
Requirement Analysis	3/4/2023	21/04/2023	19
Interface Design Layout	26/04/2023	30/05/2023	35
Design	1/8/2023	25/08/2023	25
Implementation	4/9/2023	10/11/2023	68
Sytem Testing	13/11/2023	18/11/2023	6
Reporting	20/11/2023	30/11/2023	11

Figure 1 Gantt chart

Chapter No 2: Functional Requirements

2.1. Introduction

Food for All is a progressive Android app to share food. This system makes a connection with donor and NGO. On this platform, the donor can easily donate the donation with multiple resisted NGOs and deliver his food or cash with needy and poor peoples. In the facility list of our app we have rider for getting donation from the destination of donor. The admin send the location to the rider and rider collect the food from donor and give to the NGOs through admin. In this way any donor can donate their material, item and food to the needy people. Through this application, overseas also contribute and help needy and poor people in any emergency situation.

2.2. Existing System

Table 1: Comparative Analysis

System Name	Donation	Google Map based Location	Rider	Add destination	Request for same route	Free	Geographical Zone
Food For All	✓	✓	✓	✓	✓	✓	Pakistan

Organizational Chart

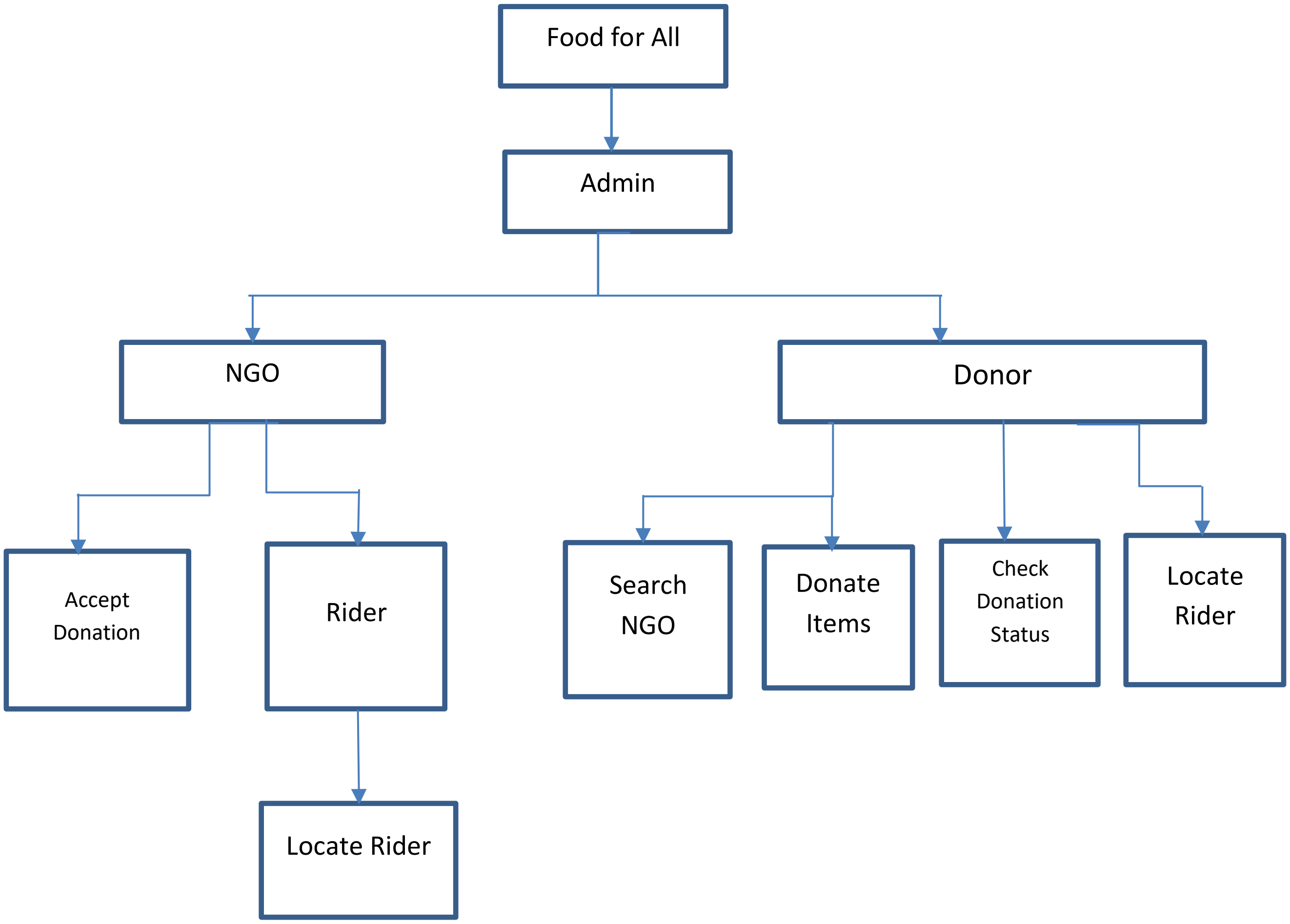


Figure 2: Organizational Chart

2.3. Scope of the System

Food for All is an android application that provides opportunities to people to donate food easily even they don't need to search a hungry one. This app allows a layman to use technology for their ease in an efficient way. Integrated maps will be used in this application which automatically gets the location of the person who wants to donate food and display the location of the person who requests for the food. Using location data, the app matches between the donor and a nearby organization to collect food. This app allows a layman to use technology for their ease in an efficient way.

Users can share the food with available Charity services. Users can find Charities near to the location and can deliver themselves, the user will get the root map assistance to a charity which will easy to reach the location. Otherwise, all the users have to upload the food images in the app and immediately all the volunteers will get a notification and will contact you to pick up the food. The user can also track the status of food till delivery. Those who ready to act as a volunteer can register as a volunteer in the app itself and can collect food from different donors and serve to Charity.

2.4. Summary of Requirements: (Initial Requirements)

The purposed system must fulfill following requirements as follow:

2.4.1 Functional Requirements

These are the requirements that the end user specifically demands as basic facilities that the system should offer. All these functionalities need to be necessarily incorporated into the system as a part of the contract.

1. **For Donor**
   a. Donate item (click on donate, then select what item and in which form you wish to donate and have a volunteer pick up your item for you)
   b. Check where the volunteer has reached and the status of your donated item
   c. Search and select NGO as per need
2. **For volunteers**
   a. Accept requests given by management. Go to the locations and pick up the items that are provided by the donor.
3. **For NGO management**
   a. Accept and manage requests by the donors and assign volunteers for pick up.
   b. Keep track of the performance and whereabouts of the volunteers
4. **For admins**
   a. Admins shall manage database information and shall do relevant tasks related to the same.

2.4.2 System Requirement

System requirements are all of the requirements at the system level that describe the functions which the system as a whole should fulfill to satisfy the stakeholder needs and requirements, and are expressed in an appropriate combination of textual statements, views, and non-functional requirements; the latter expressing the levels of safety, security, reliability, etc., that will be necessary.

System requirements play major roles in systems engineering, as they:

- Form the basis of system architecture and design activities.
- Form the basis of system integration and verification activities.
- Act as reference for validation and stakeholder acceptance.
- Provide a means of communication between the various technical staff that interact throughout the project.

2.5 Identifying External Entities

The Identification of External Interfaces is done in two phases.
Over Specify Entities from Abstract:

System Admin
Users/donor

2.6 Capture "shall" statements

Table 2 : Initial Requirements

Para #	Initial Requirements
1.0	The system "shall" provide different types of registration, admin, donor, rider
1.0	System "shall" update the user's request
1.0	System Admin "shall" view registrations.
1.0	System "shall" accept, reject the request
1.0	User shall register himself as donor
1.0	User "shall" search available NGOs
1.0	A user "shall" complaint (if required)
1.0	A user "shall" login to the system and can change his password

2.7 Allocate Requirements

Table 3: Requirements Allocation

Para#	Initial Requirements	Use Case Name
1.0	User shall register himself as Donor	UC_Registration_Request
1.0	System Admin “shall” view registrations	UC_View_ registration
1.0	System “shall” accept, reject the request	UC_Accept/Reject _User_Request
1.0	User “shall” search NGOs	UC_Search_Location
1.0	A User “will” place Request for Donation	UC_Place_Request
1.0	A User “will” edit Request for Donation	UC_Edit_Request
1.0	System “shall” update his profile	UC_Update_Profile

2.8 Prioritize Requirements

Table 4: Requirements Priorities

Para#	Rank	Initial Requirements	Use Case ID	Use Case Name
1.0	High	User shall register himself as Donor	UC_01	UC_Registration_Request
1.0	Medium	System Admin “shall” view registration.	UC_02	UC_View_ registration
1.0	High	System “shall” accept or reject the request	UC_03	UC_Accept/Reject_User_Request
1.0	High	A User “shall” login to the system	UC_04	UC_Login
1.0	Medium	User “shall” search rider Location	UC_06	UC_Search_Location
1.0	Medium	A User “will” place request fordonation	UC_07	UC_Place_Request

1.0	Low	A User “will” edit request fordonation	UC_08	UC_Edit_Request
1.0	Medium	System “shall” update his profile	UC_10	UC_Update_Profile
2.0	Low	User “shall” view the status ofrider	UC_11	UC_View_Booked_Ride
3.0	Medium	System Admin “shall” view his action list containing different actions to be done	UC_13	UC_View_Action

Chapter No 3: Non- Functional Requirements

3.1 Non-functional Requirements

These are basically the quality constraints that the system must satisfy according to the project contract. They are also called non-behavioral requirements.

They basically deal with issues like:

1. **Portability**
   Since this is an android application, it shall be available for the majority of android users on their mobile phones.
2. **Usability**
   The system needs to be easy to use and understand. Our application will be easy for users to navigate through and they can easily use the functionalities of the app in a smooth manner. Changing of various activities shall allow for easy shifting of tasks.
3. **Privacy**
   The application shall take care of not to leak any donor's personal profile information to other users or people. Only information relevant and necessary shall be visible.
4. **Performance**
   The app should respond to users in a considerable time window. It should not be too slow or too fast for the users. The application's response time should not hinder the user in his tasks.
5. **Scalability**
   App should able to adapt itself to the increased user load or more users in order to handle more data as time progresses.
6. **Reliability**
   The application should be reliable to perform its tasks. For example, a user should rely that in case he wishes to donate an item, request is sent to the volunteer side otherwise it would be an unreliable app.

Chapter No 4: Assumption & Dependencies

4.1 Task Dependency Table

The following are the steps to develop a task dependency table:

- 7th Semester

Table 5: Task Dependency 1

Tasks	Implementation Time
Proposal	Started in first month and completed in the same month.
Requirement Gathering	It also started in first month and completed within the 1st month.
Requirement Analysis	It started in second month.
Interface Design layout	It starts in second month and completed before this semester.

- 8th Semester

Table 6: Task Dependency 2

Tasks	Implementation Time
Design	It will be start in 8th Semester and complete in one month.
Implementation	It will start after the design and complete in 3 months.
System Testing	It will take 6 days
Reporting	After completing implementation and testing phase report will be complete within 10 days.

4.2 CPM-Critical Path Method

1. Specify the Individual Activities

Following are the individual activities involved in the project:

Start

A) Communication

B) Planning

C) Modeling

D) Construction

E) Deployment

2. Determine the Sequence of the Activities

Start None

A) Communication → Start

B) Planning → Communication

C) Modeling → Planning, Communication

D) Construction → Modeling, Planning

E) Deployment → Implementation

F) End → Deployment

Table 7: Task Description and Duration

Task ID	Task Description	Duration (Days)	Dependencies
-	Start	-	-
A	Communication	14	-
B	Planning	19	A
C	Modeling	60	A,B
D	Implementation	68	B,C
E	Deployment	17	D

3. Draw the Network Diagram

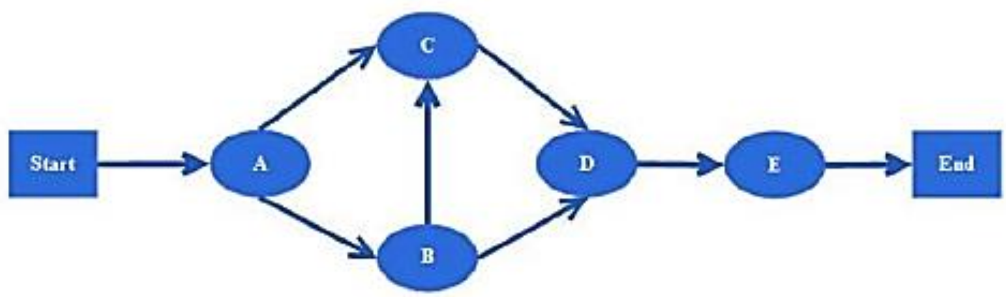


Figure 3: Network Diagram

4. Estimate Activity Completion Time

Task ID	Predecessors	Duration (Days)
A	-	14
B	A	19
C	A,B	60
D	B,C	68
E	D	17

5. Identify the Critical Path

The Critical path is A, B, C, D, E = 14+19+60+68+17 = 178 days.

6. Update CPM Diagram

A → B → C → D → E

Table 8: Critical Path

Activity	Duration	ES	EF	LS	LF	TS	FS
A	14	0	15	0	15	0	0
B	19	15	30	15	30	0	0
C	60	15	45	15	45	0	0
D	68	30	70	45	95	0	0
E	17	70	90	95	115	0	0

4.3 Introduction to team members and their skills set

Table 9: Introduction to team members and their skills

Members Name	Members Roll No	Skill Set
Aneesa Iftikhar	19BSCS29213	Database Design & Development, Android Front end designing and coding.
Ayeza Akram	19BSCS29211	Backend Coding, Documentation and Database Designing.
Ayeza Ijaz	19BSCS29209	Front end Designing and Documentation

4.4 Task and Member Assignment Table

Table 10: Task and Member Assignment

Task	Duration (days)	Dependencies	Members
T1	14		M1, M2,M3
T2	19	T1	M1, M2,M3
T3	35	T1, T2	M1, M2
T4	25	T1, T2, T3	M1, M2
T5	68	T1, T2, T3, T4	M1, M2,M3

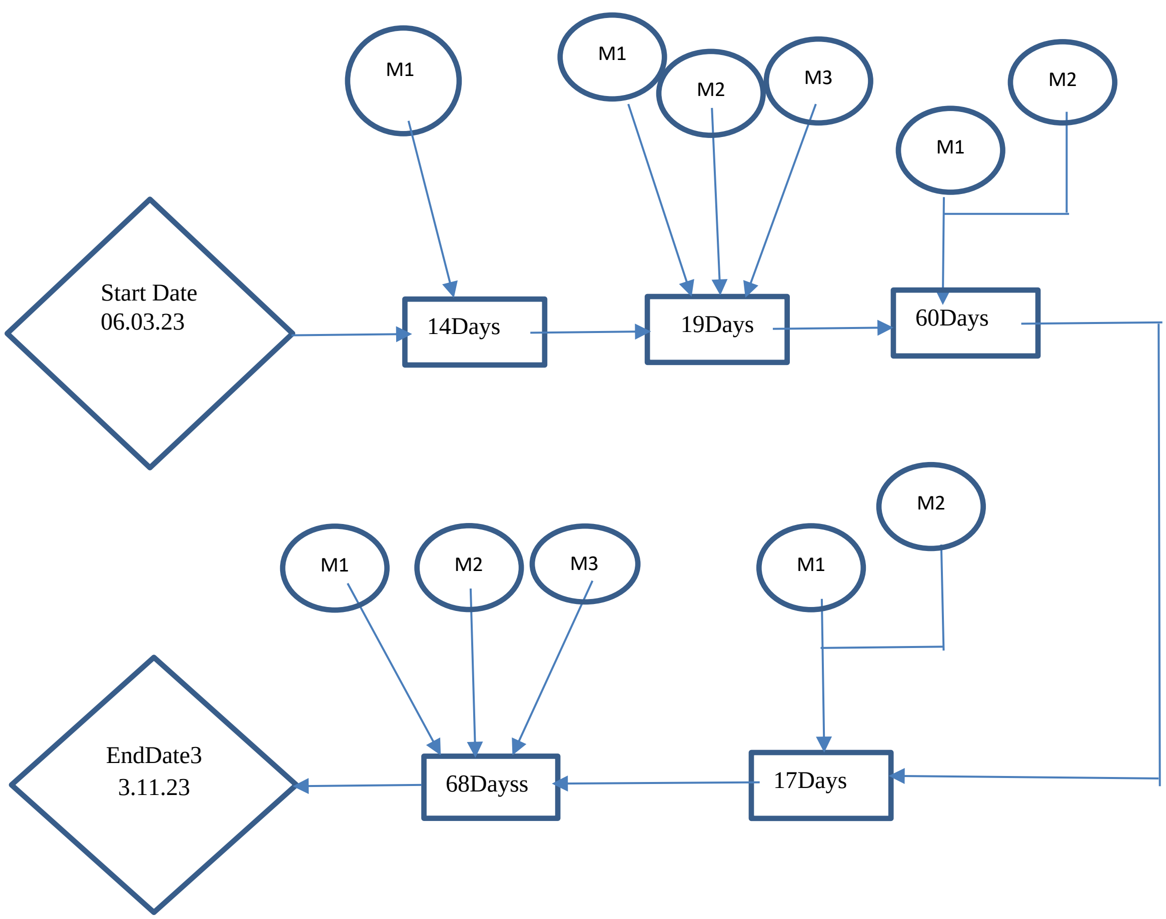


Figure 4: Task and Member Assignment Diagram

Task duration and Dependencies

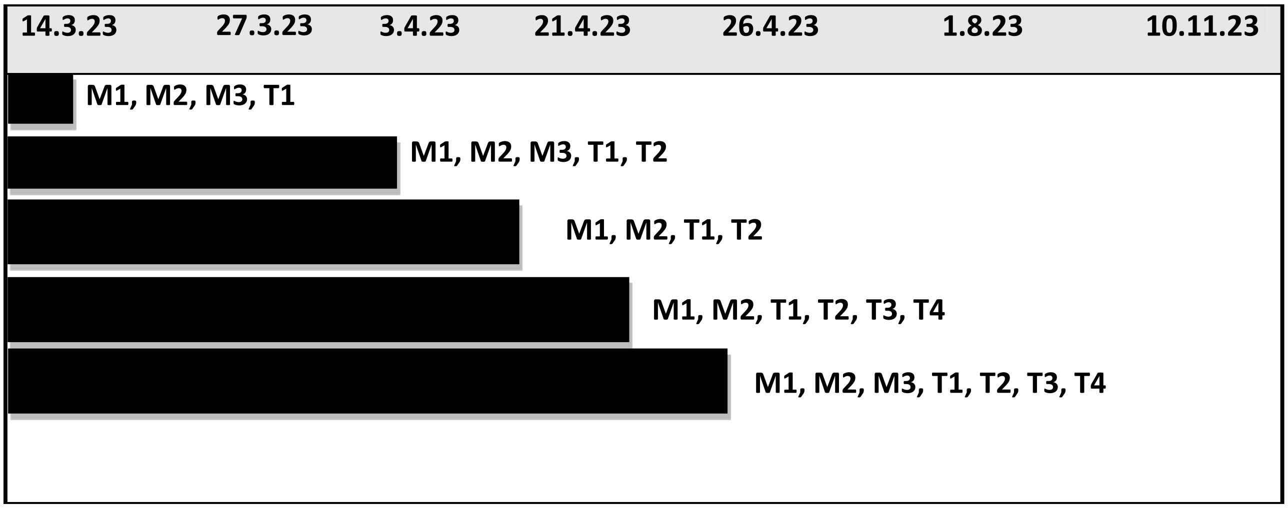


Figure 5: Task Duration and Dependencies

Allocation of People to Activities:

Table 11: Allocation of people to Activities

Task	Member
T1	**Aneesa Iftikhar**
T2	**Ayeza Akram**
T3	**Aneesa Iftikhar**
T4	**Ayeza Akram**
T5	**Ayeza Ijaz**
T6	**Ayeza Ijaz**
T7	**Aneesa Iftikhar**
T8	**Ayeza Akram**
T9	**Aneesa Iftikhar**
T10	**Ayeza Akram**

Activity Bar chart

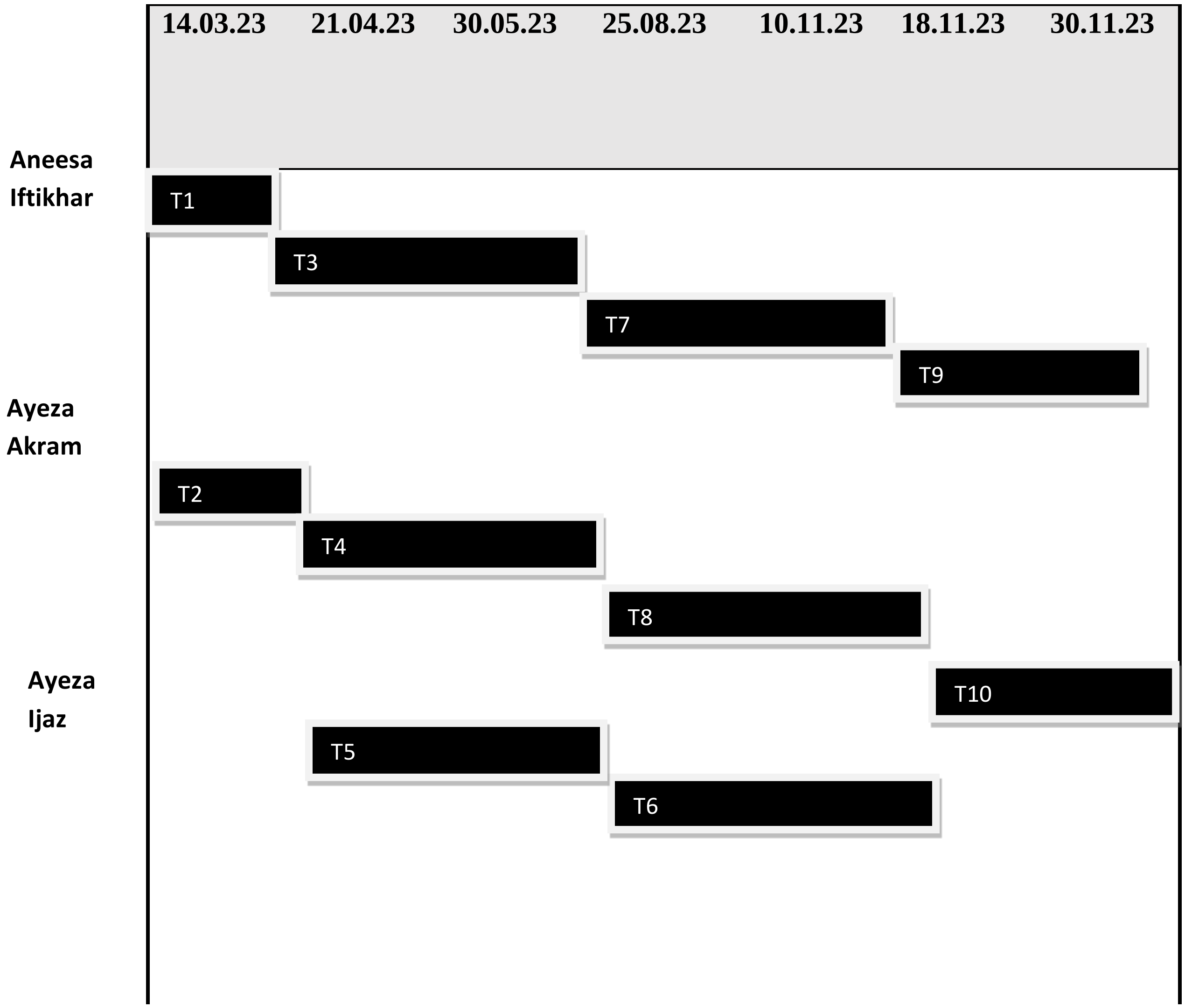


Figure 6: Activity Bar chart

Chapter No 5: System Architecture

5.1 Context level data flow diagram

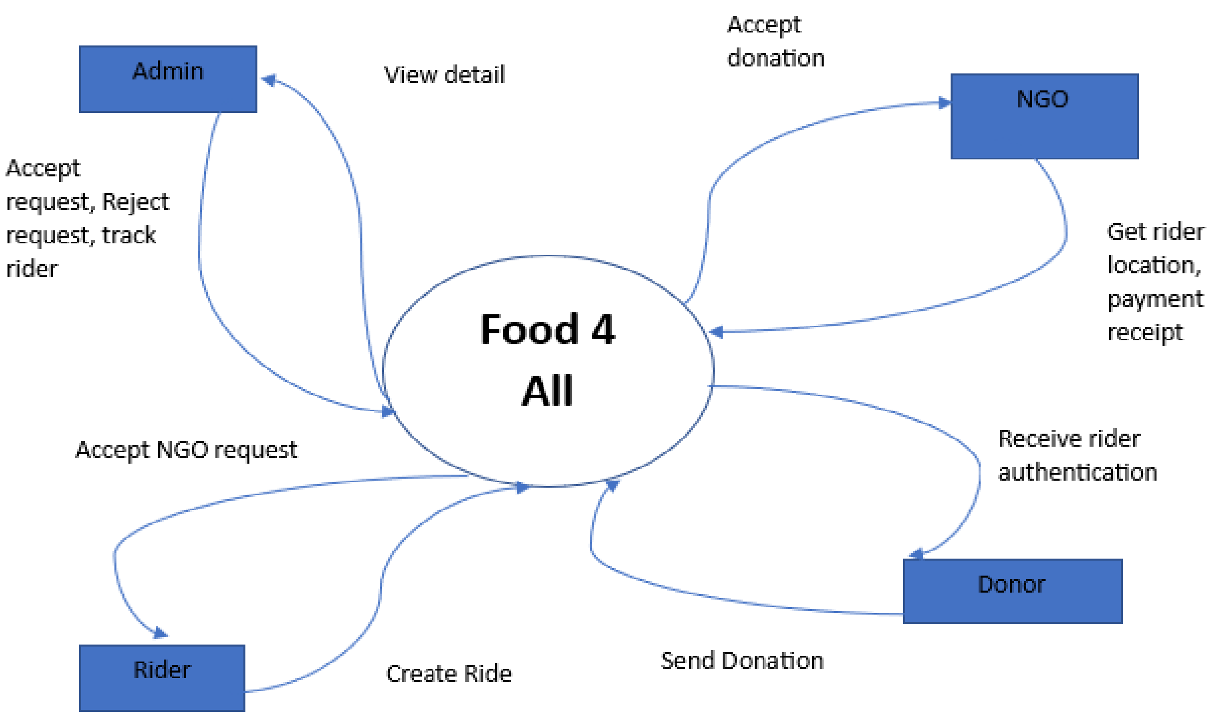


Figure 7: DFD Diagram

Chapter No 6: Use Cases

6.1 High Level Use case Diagram:

Figure 8: High level use case Diagram

6.2 Analysis Level Use case Diagram:

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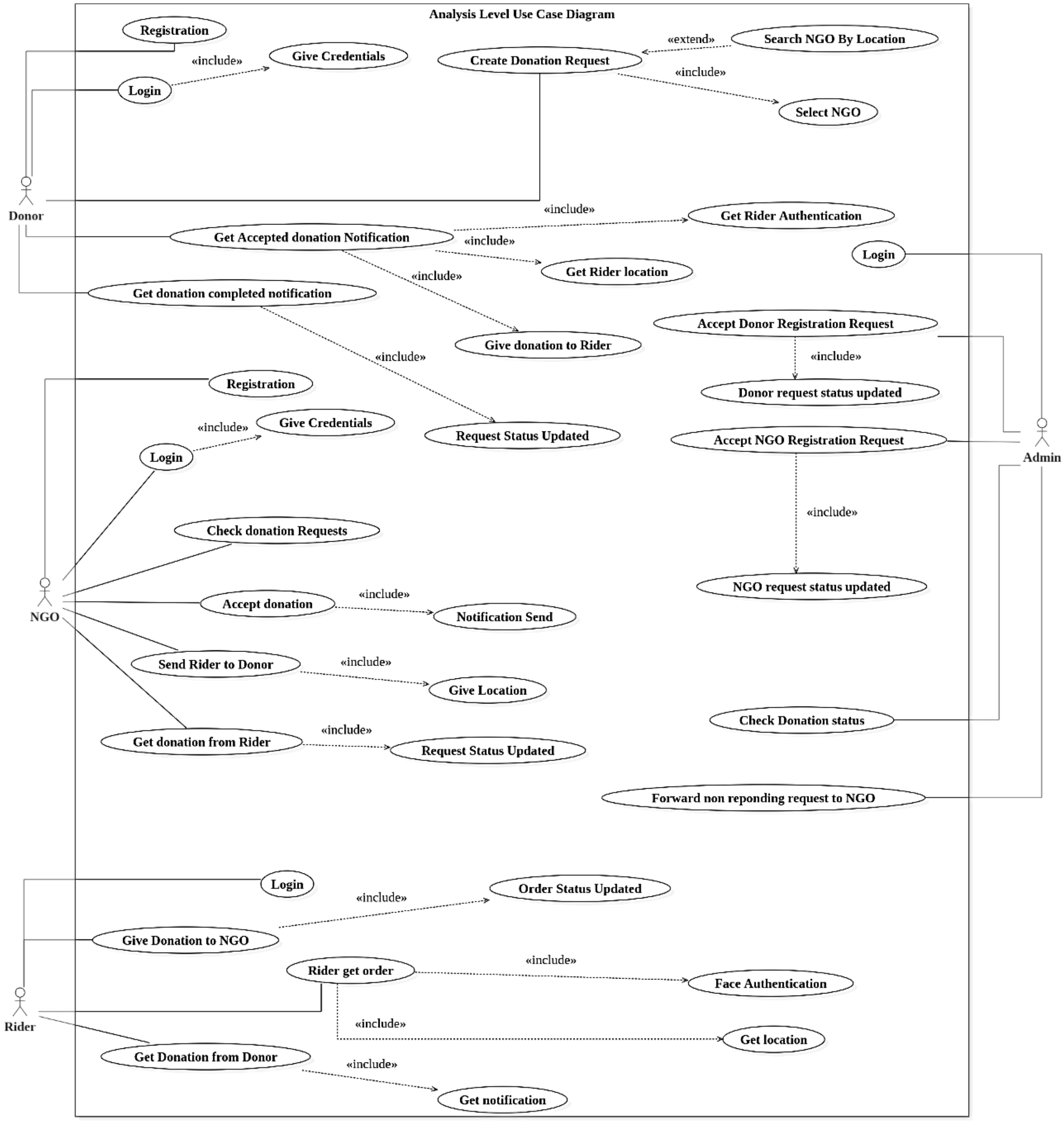


Figure 9: Analysis level use Case Diagram

6.3 Use case Description

Use case Description

In the description of use case document contain the following parts:

UC_1:

Name	Food For All
Scope	it is mobile application, every person can use it such as foreign, and organizations
Level	Admin system, mobile
Stack holder and interest	Admin can manage application
Brief description	In this use case primary actor “Donor” will register himself in the application and check status
Precondition	Visitor must visit the application.
Basic flow	Donor register himself for donation and check the list of registered NGOs.
Alternate flow	The Admin will not show the unregistered NGOs
Post conditions	Admin has successfully show the NGOs list.

UC_2:

Brief description	In this use case primary actor “Donor” shall make the login and the secondary actor “Admin” approve the donor login.
Precondition	Admin or Donor have login.
Basic flow	Admin check the donor detail after login of donor
Alternate flow	The Admin will not show the donor list to the admin due to connection failure
Post conditions	Admin has successfully show the list.
Name	Food For All
Scope	it is mobile application, every person can use it such as foreign, and organizations
Level	donor system, mobile
Stack holder and interest	Donor can donate

UC_3:

Brief description	In this use case primary actor “Donor” give the donations request and secondary actor “admin” check the request of donor
Precondition	The donor must have a donation request.
Basic flow	The donor generate the donation request and the admin approve this.
Stack holder and interest	Donor generate request, admin accept the reqest
Scope	it is mobile application, every person can use it such as foreign, and organizations
Level	Admin system, mobile

UC_4:

Brief description	In this use case primary actor “Donor” after the request the donor check the status of rider for collection of donation.
Precondition	The donor must send the request.
Basic flow	Donor send his request and the admin allocate the rider for collection of donation.
Scope	it is mobile application, every person can use it such as organizations
Level	Rider system, mobile
Stack holder and interest	Donor can check donation request and rider receive the request

UC_5:

Brief description	In this use case the primary actor “NGO” will login to the Admin and then check the donation detail while the secondary actor “Admin” will save this addition in to the database of the Admin.
Precondition	“NGO” must visit the applcation.
Basic flow	NGO register himself in the Admin and admin allow to access to the donator
Alternate flow	The Admin will not save the NGO into his database due to any reason.
Post conditions	Admin has successfully add the NGO save into his database.
Scope	it is mobile application, every person can use it such as foreign, and organizations
Level	NGo system, mobile
Stack holder and interest	NGO can check all donation requests

UC_6:

Brief description	In this use case the primary actor “NGO” will login to the Admin and check the donation detail if it agreed. They send the rider on the donation destination for collection of donation
Precondition	“NGO” must visit the App.
Alternate flow	The Admin will not save the NGO into his database due to any reason.
Post conditions	Admin has successfully add the NGO save into his database.

UC_7:

Brief description	In this use case the primary actor “NGO” receive the donation from rider and update the status and “Admin” will update the status into the database of the Admin.
Precondition	“Rider” must visit the App.
Alternate flow	The Admin will not save the Donation detail into his database due to any reason.
Post conditions	Admin has successfully add the Donation detail save into his database.
Scope	it is mobile application, every person can use it such as organizations
Level	NGO system, mobile
Stack holder and interest	NGO receive all the updates

UC_8:

Brief description	In this use case the primary actor “Rider” have login in the app and “Admin” approve his login in the Admin.
Precondition	“rider” registration is pending from Admin
Alternate flow	The Admin will not save the rider into his database due to any reason.
Post conditions	Admin has successfully add the rider save into his database.
Scope	it is mobile application, every person can use it such as organizations
Level	Rider system, mobile
Stack holder and interest	Rider receive pickup request

UC_9:

Brief description	In this use case the primary actor “rider” get the order from NGO and update the status and “Admin” will update the status into the database of the Admin.
Precondition	“Rider” must visit the App
Alternate flow	The Admin will not save the rider detail into his database due to any reason.
Post conditions	Admin has successfully add the rider detail save into his database.

Scope	it is mobile application, every person can use it such as organizations
Level	rider system, mobile
Stack holder and interest	Rider get the order from NGO

UC_10:

Brief description	In this use case the primary actor “Rider” receive the donation from Donor and update the status and “Admin” will update the status into the database of the Admin.
Precondition	“rider” must visit the App.
Alternate flow	The Admin will not save the Donation into his database due to any reason.
Post conditions	Admin has successfully add the Donation save into his database.

UC_11:

Brief description	In this use case the primary actor “rider” deliver the donation in NGO and update the status and “Admin” will update the status into the database of the Admin.
Precondition	“rider” must visit the App.
Alternate flow	The Admin will not save the Donation detail into his database due to any reason.
Post conditions	Admin has successfully add the Donation detail save into his database.
Scope	it is mobile application, every person can use it such organizations
Level	Rider system, mobile
Stack holder and interest	Rider deliver the donation to NGO

UC_12:

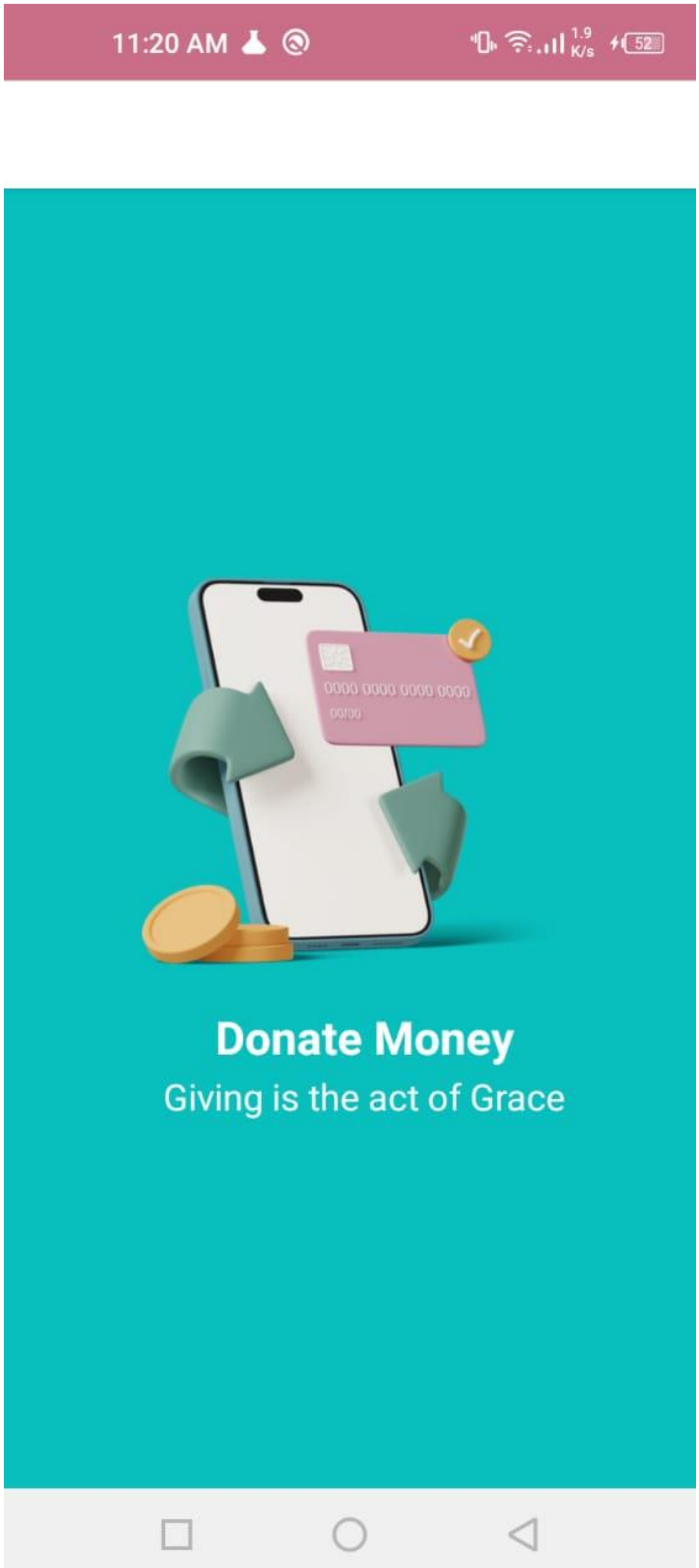
Brief description	In this use case the primary actor “GUEST” makes a login and “Admin” updates this
Precondition	“Guest” must visit the App.
Alternate flow	The Admin will not save the Guest detail into his database due to any reason.
Post conditions	Admin has successfully add the Guest detail save into his database.
Scope	it is mobile application, every person can use it such as foreign, and organizations
Level	guest system, mobile
Stack holder and interest	Guest can visit the app and donate

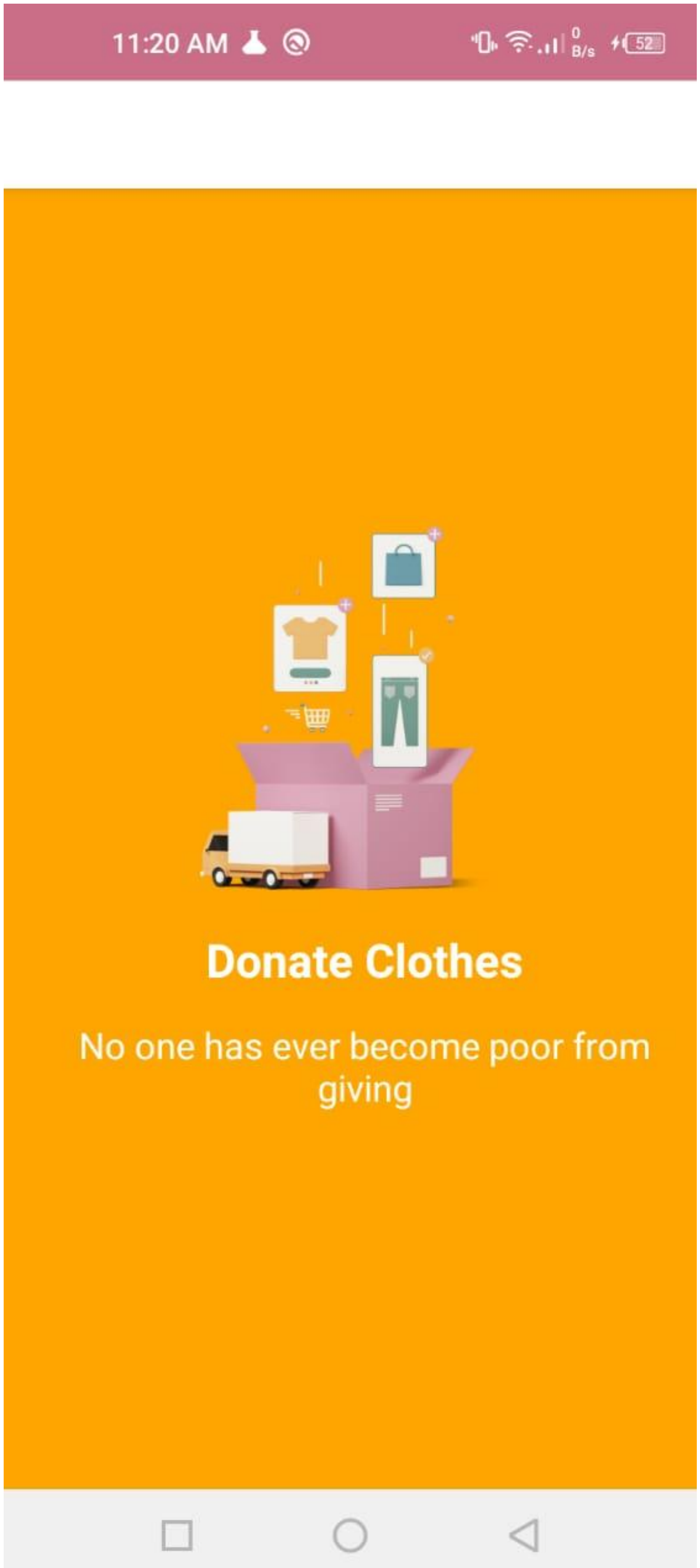
UC_13:

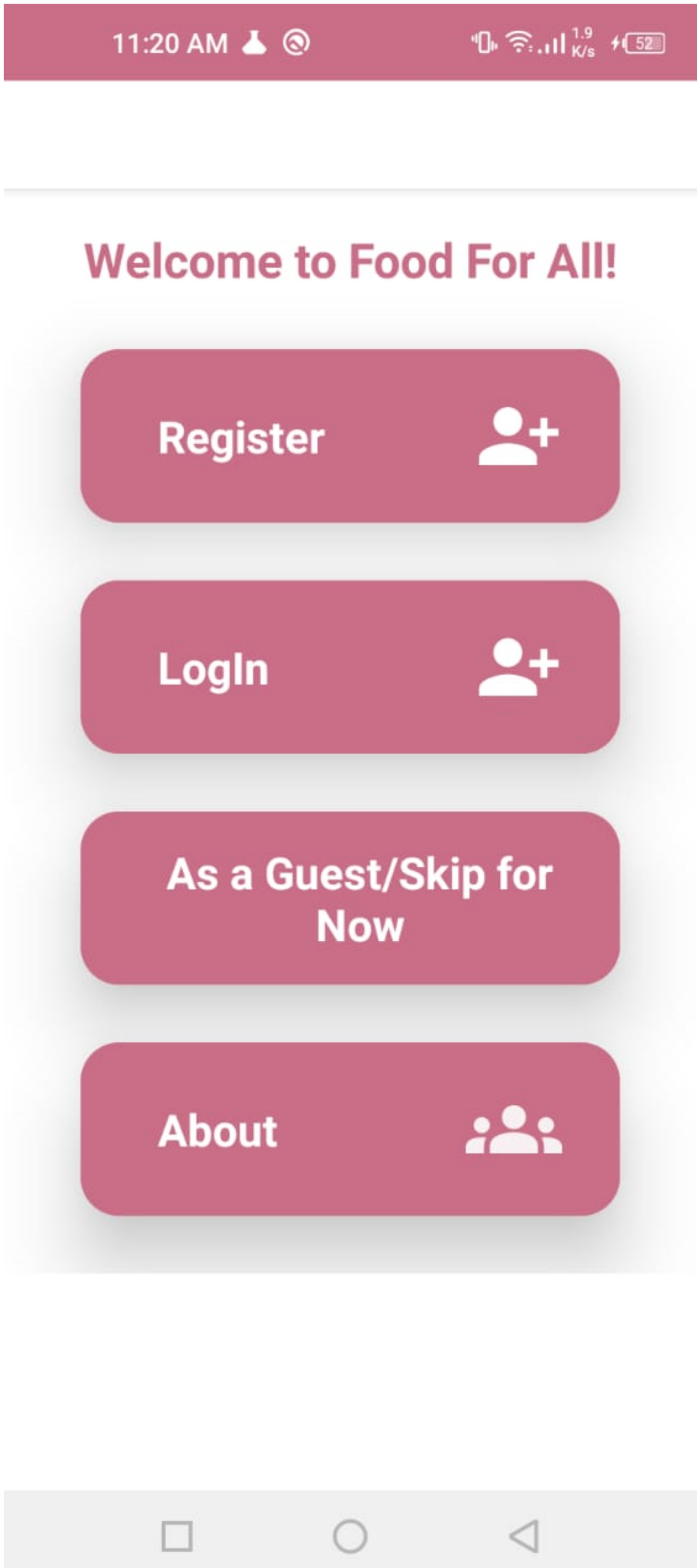
Brief description	In this use case the primary actor “Guest” give the location.
Precondition	“Guest” must visit the App.
Alternate flow	The Admin will not save the Guest into his database due to any reason.
Post conditions	Admin has successfully add the Guest save into his database.
Scope	it is mobile application, every person can use it such as foreign, and organizations
Level	Guest system, mobile
Stack holder and interest	Guest visit the app and can donate

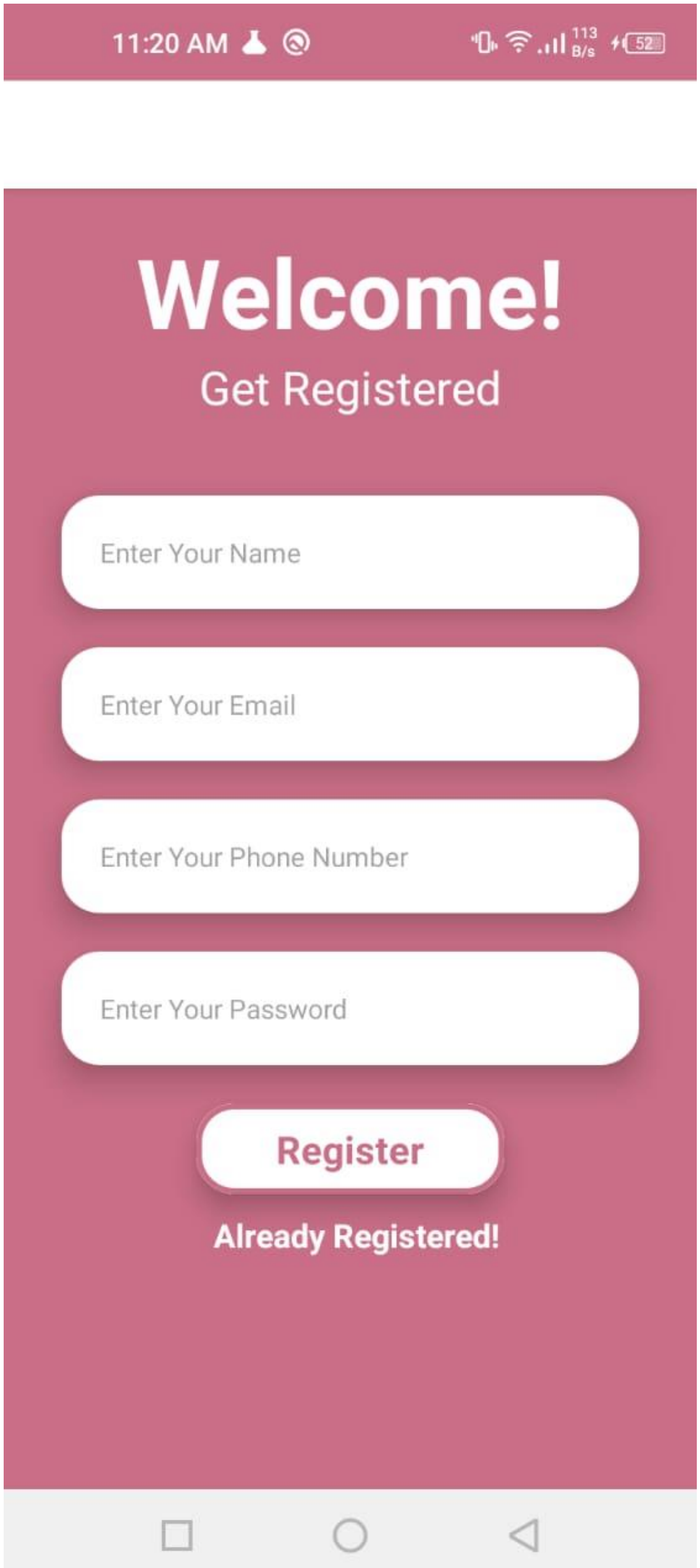
Chapter No 7: Graphical User Interface

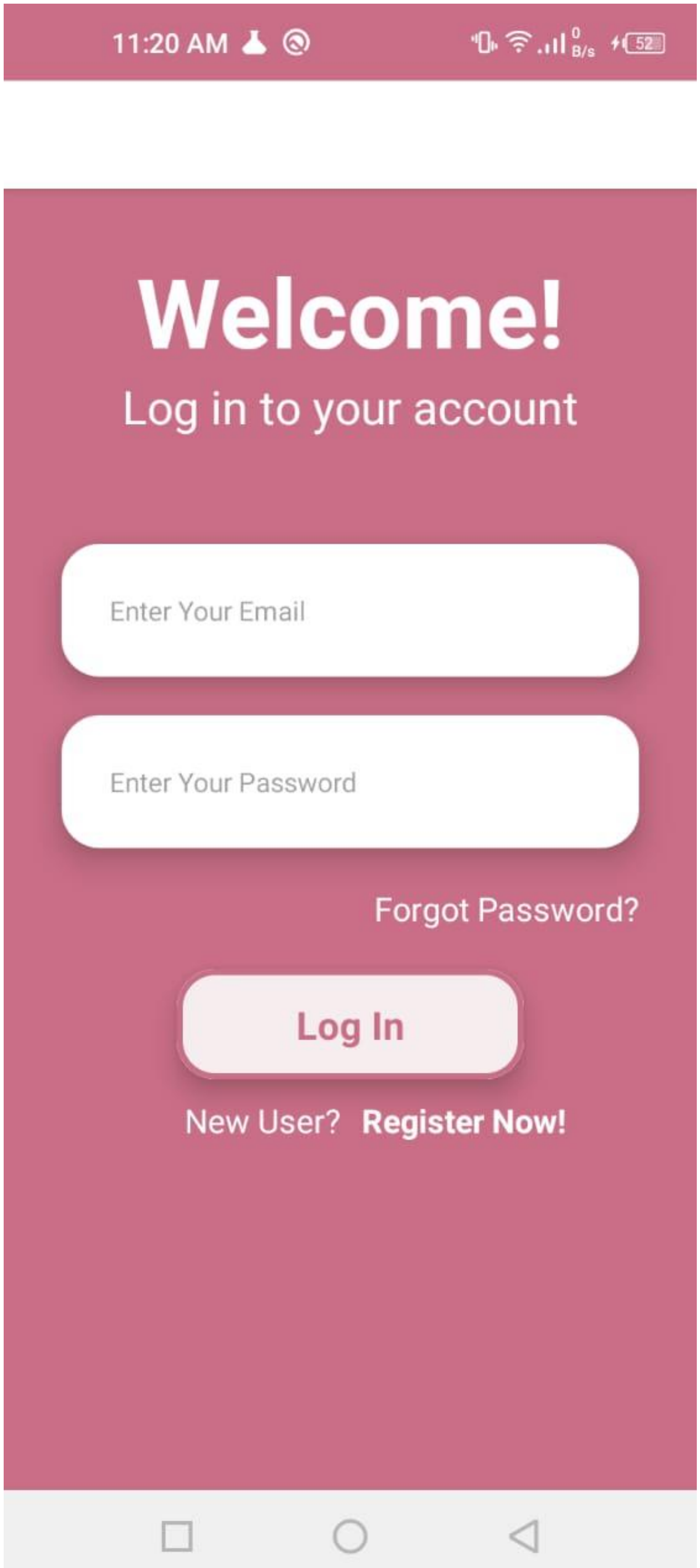

11:19 AM
Donate Food
Alone we can do so little; Together we can do so much.

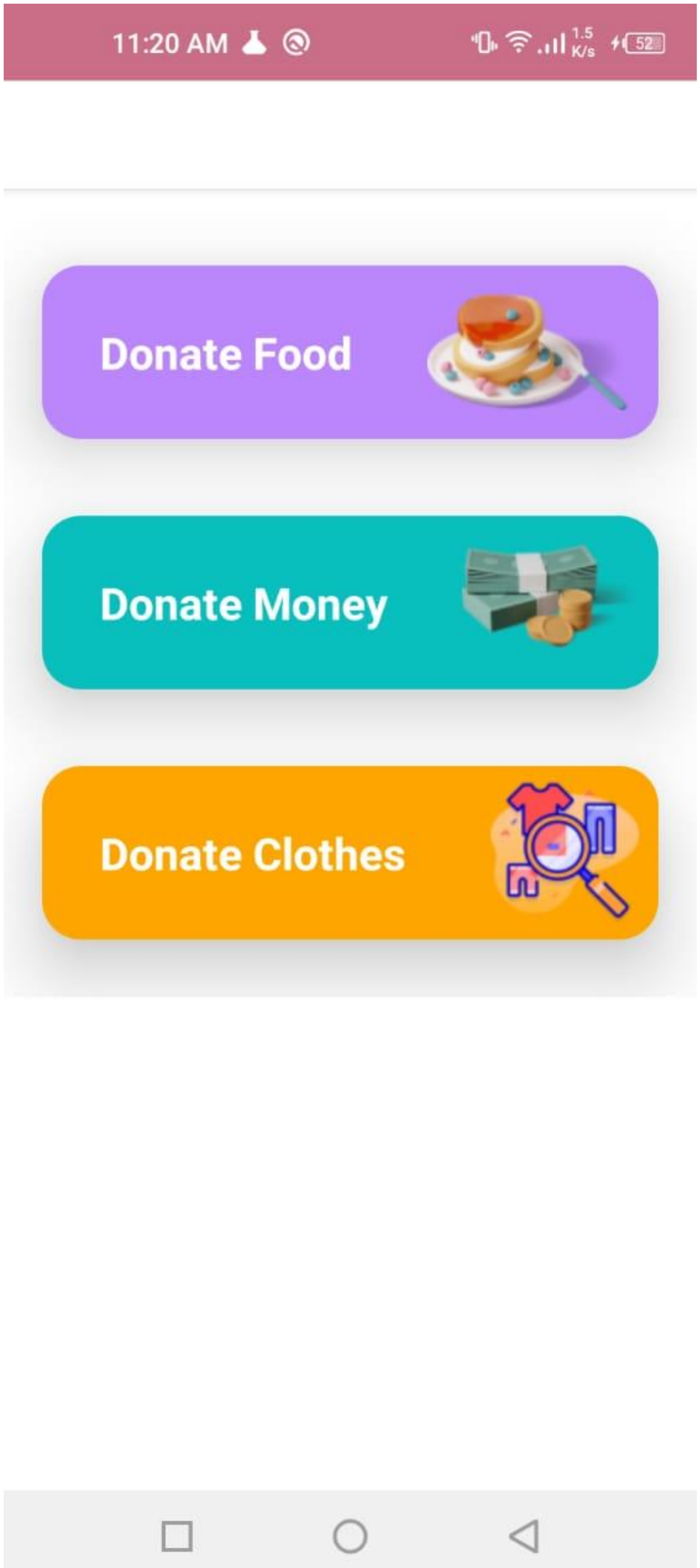

11:20 AM
Donate Money
Giving is the act of Grace

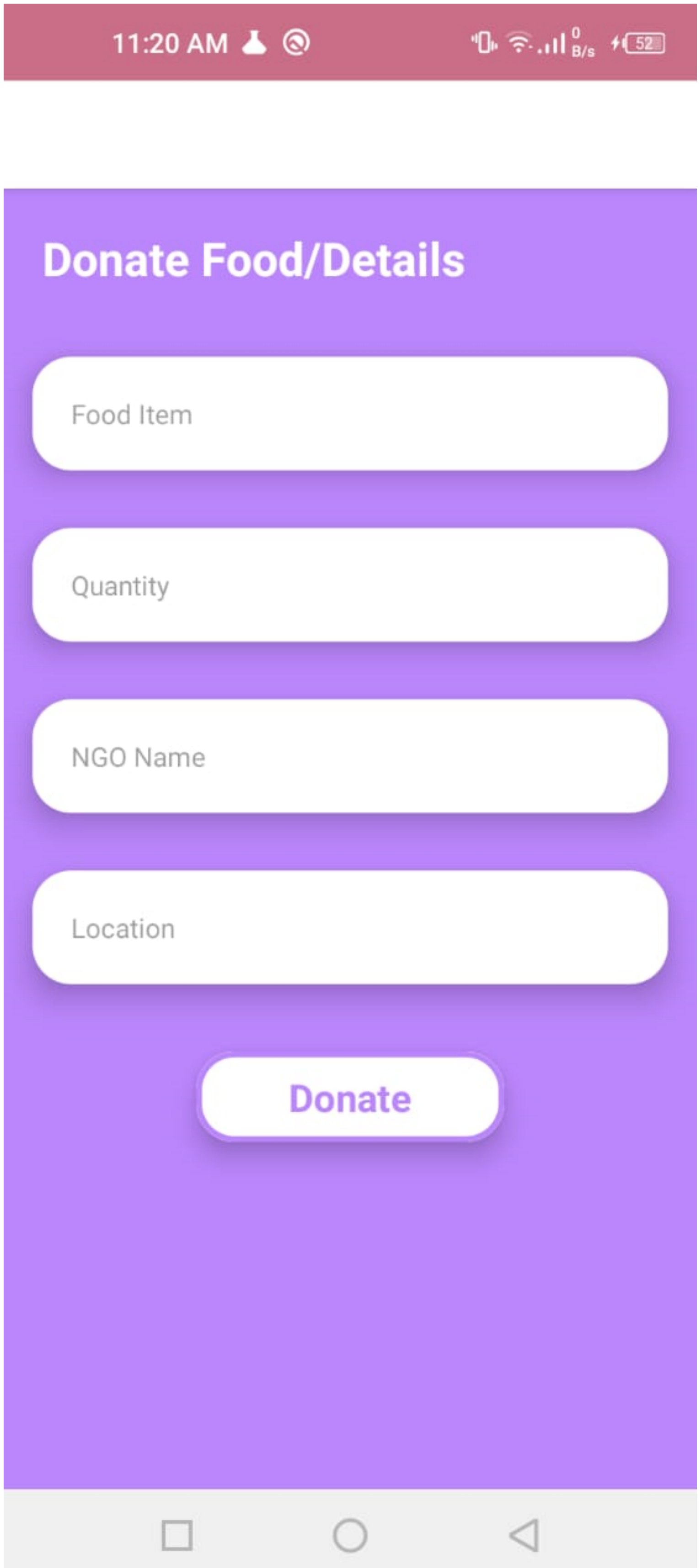

11:20 AM
Donate Clothes
No one has ever become poor from giving

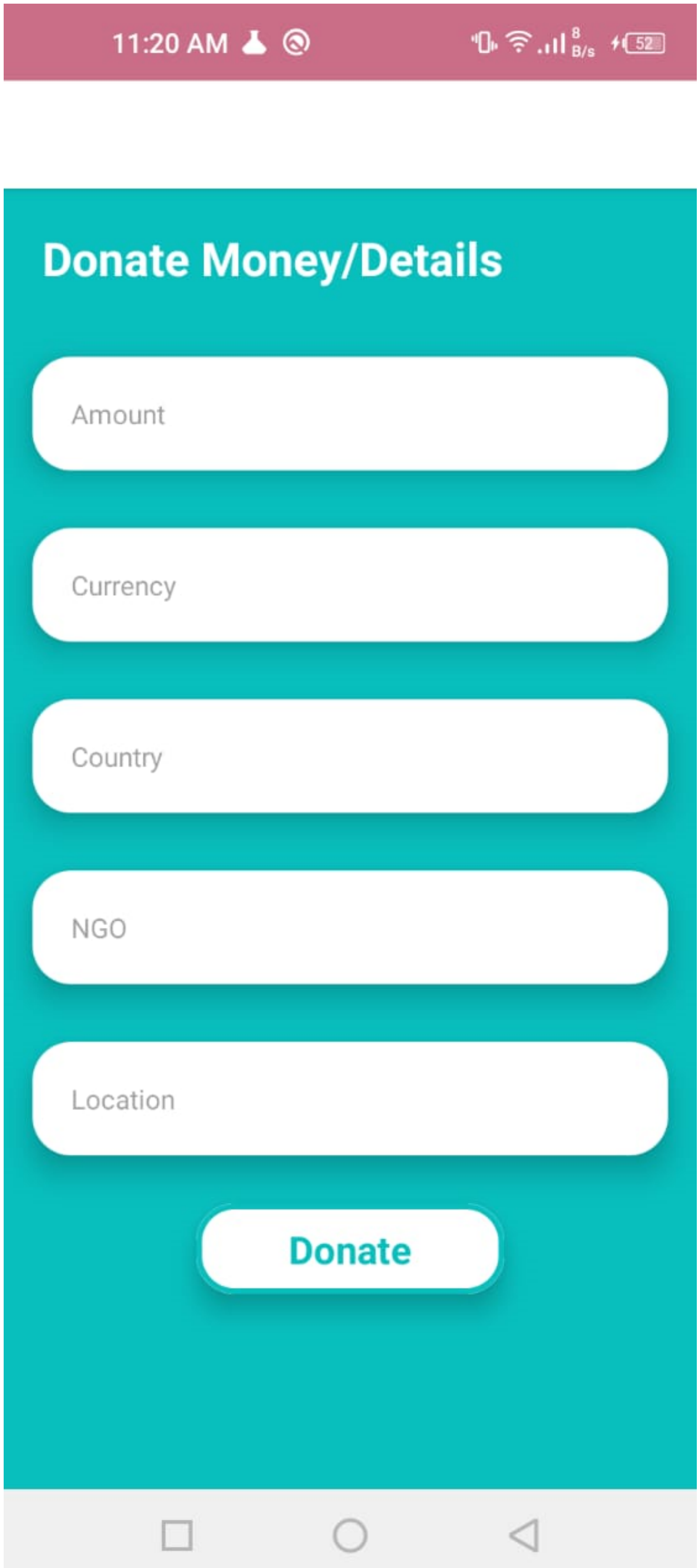

11:20 AM
Welcome to Food For All!
Register
LogIn
As a Guest/Skip for Now
About

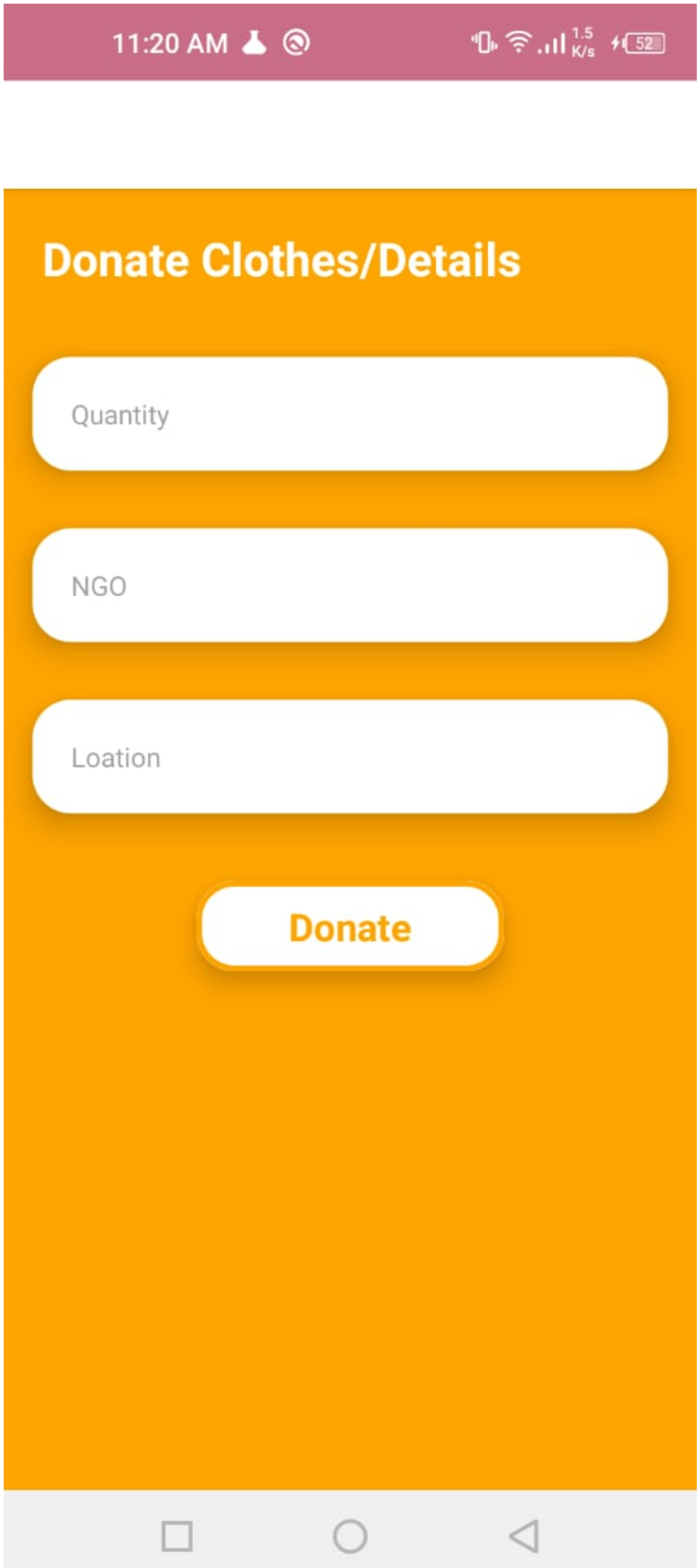

11:20 AM
Welcome!
Get Registered
Enter Your Name
Enter Your Email
Enter Your Phone Number
Enter Your Password
Register
Already Registered!


11:20 AM
52
Welcome!
Log in to your account
Enter Your Email
Enter Your Password
Forgot Password?
Log In
New User? Register Now!


11:20 AM
Donate Food
Donate Money
Donate Clothes


11:20 AM
Donate Food/Details
Food Item
Quantity
NGO Name
Location
Donate


11:20 AM
Donate Money/Details
Amount
Currency
Country
NGO
Location
Donate


11:20 AM
Donate Clothes/Details
Quantity
NGO
Loation
Donate

Chapter 8: Diagrams

8.1 Introduction:

Third deliverable is all about the software design. In the previous deliverable, analysis of the system is completed. So we understand the current situation of the problem domain. Now we are ready to strive for a solution for the problem domain by using object-oriented approach. Following modules are required to develop and implement the system for safely deployment.

1. Domain Model
2. System Sequence Diagram
3. Sequence Diagram
4. Collaboration Diagram
5. Operation Contracts
6. Design Class Diagram
7. State Transition Diagram
8. Data Model

Now we discuss these artifacts one by one as follows:

8.2 Domain Model

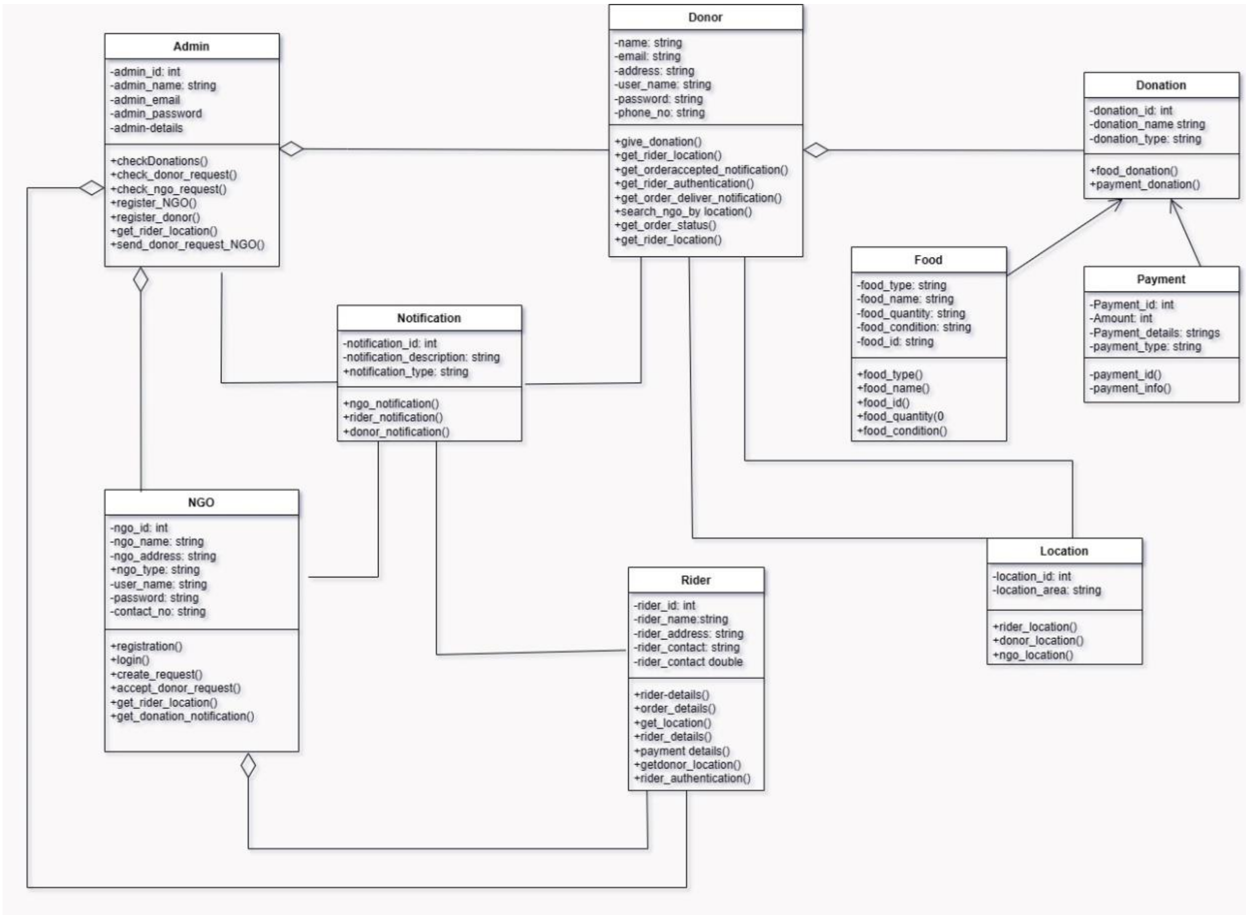


Figure 10: Domain Model

8.3 System Sequence Diagram

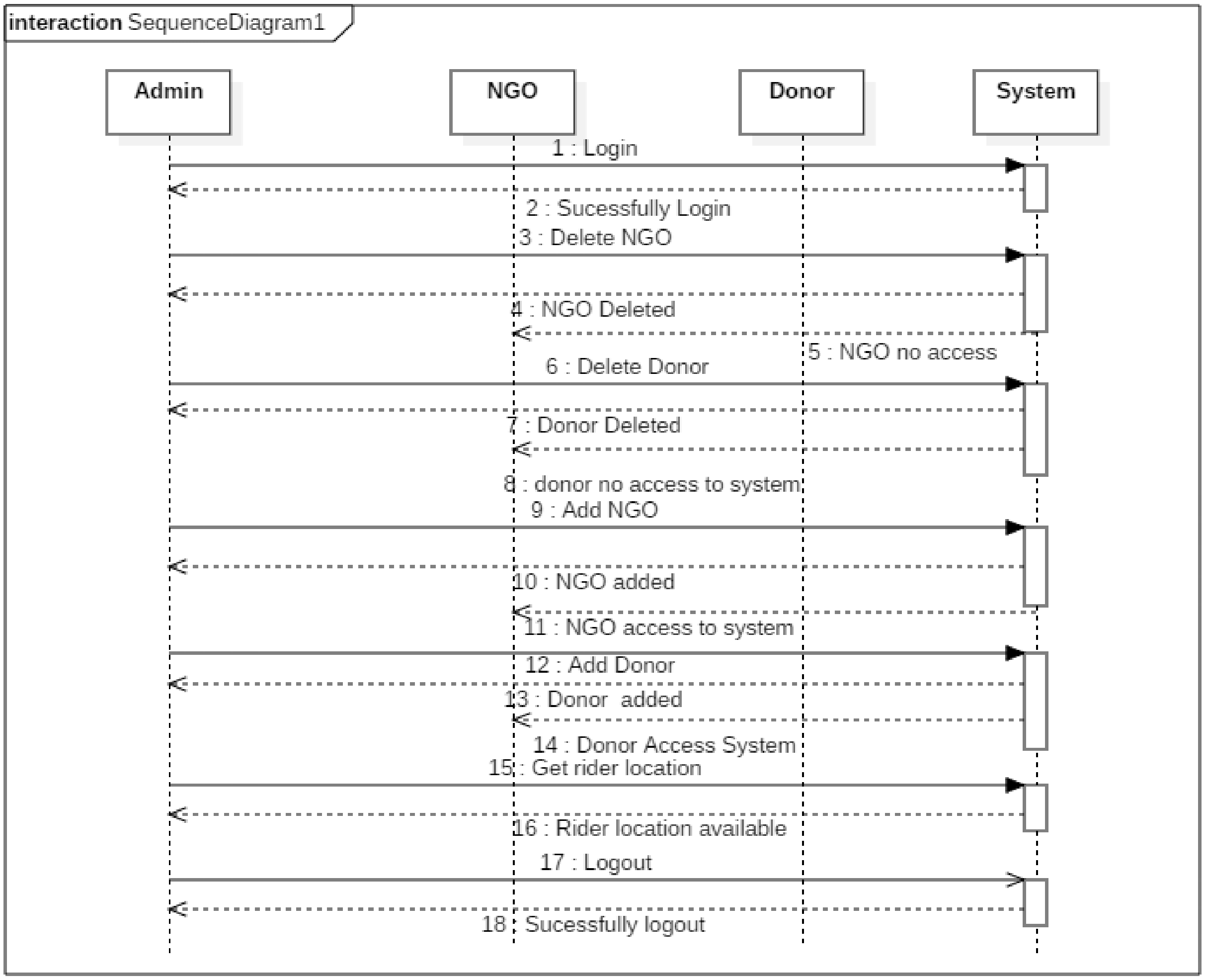


Figure 11: System Sequence Diagram

8.4. Sequence Diagram

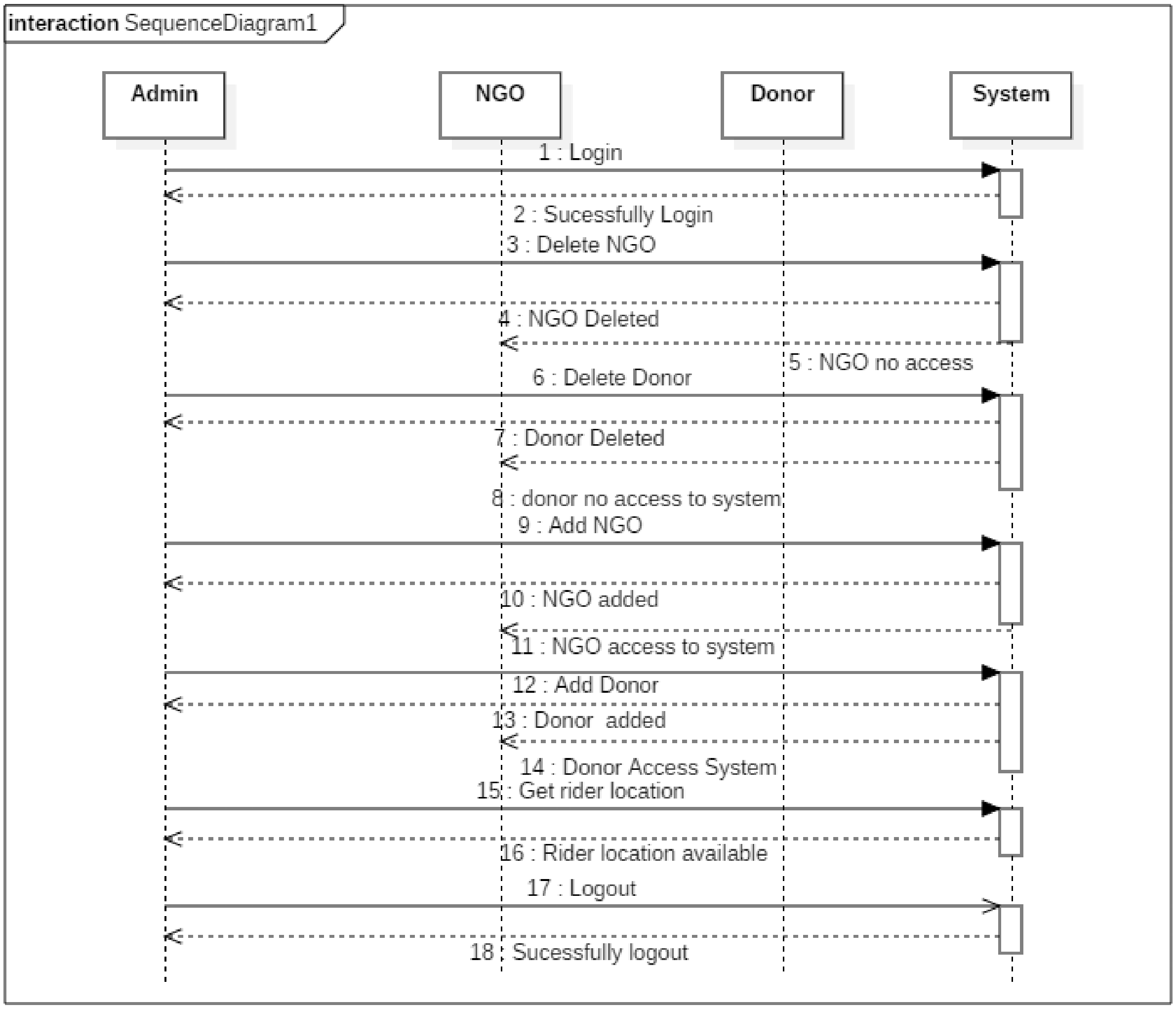


Figure 12: Admin Sequence Diagram

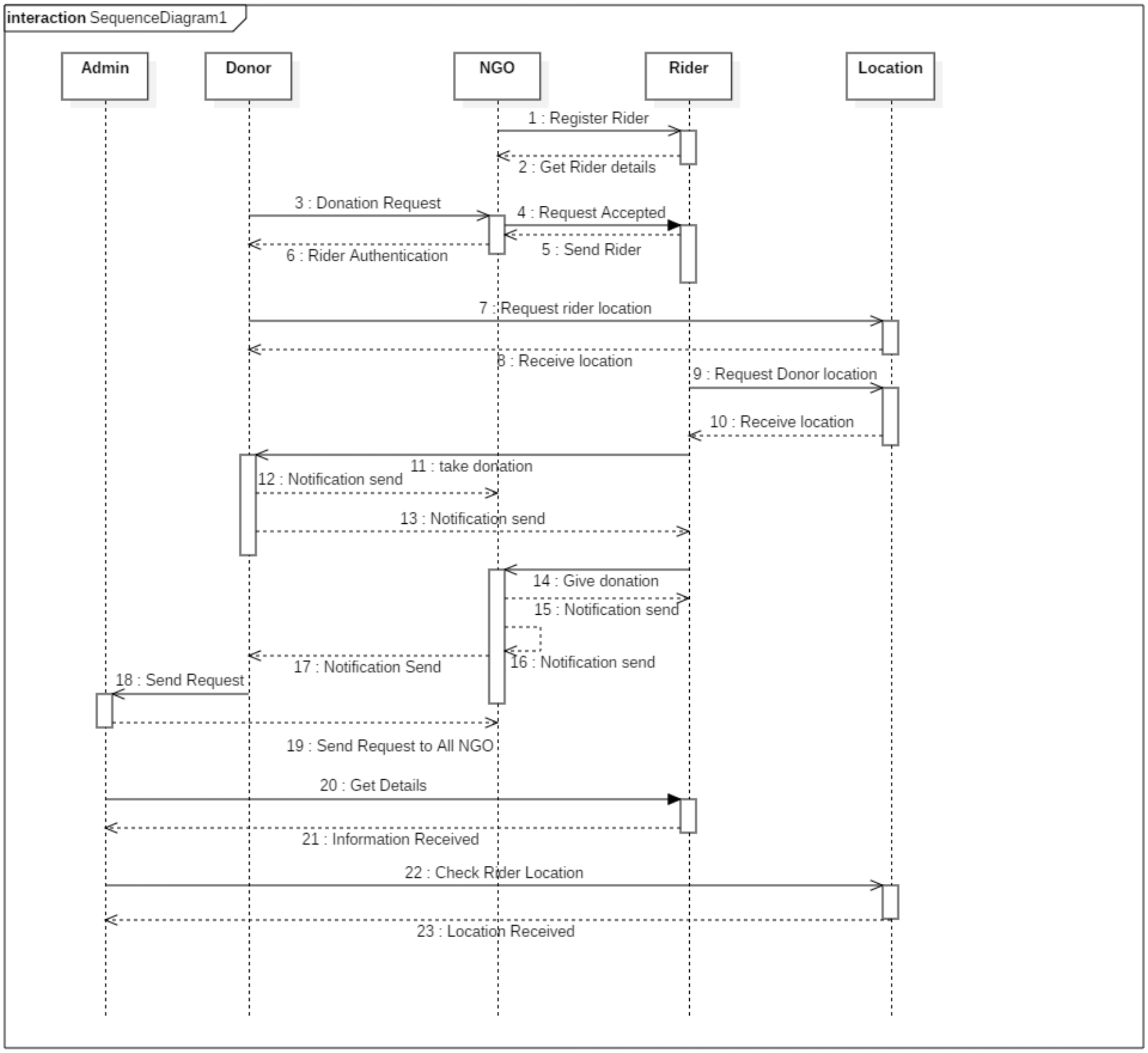


Figure 13: Admin, Rider and Donor Sequence Diagram

8.5. Collaboration Diagram

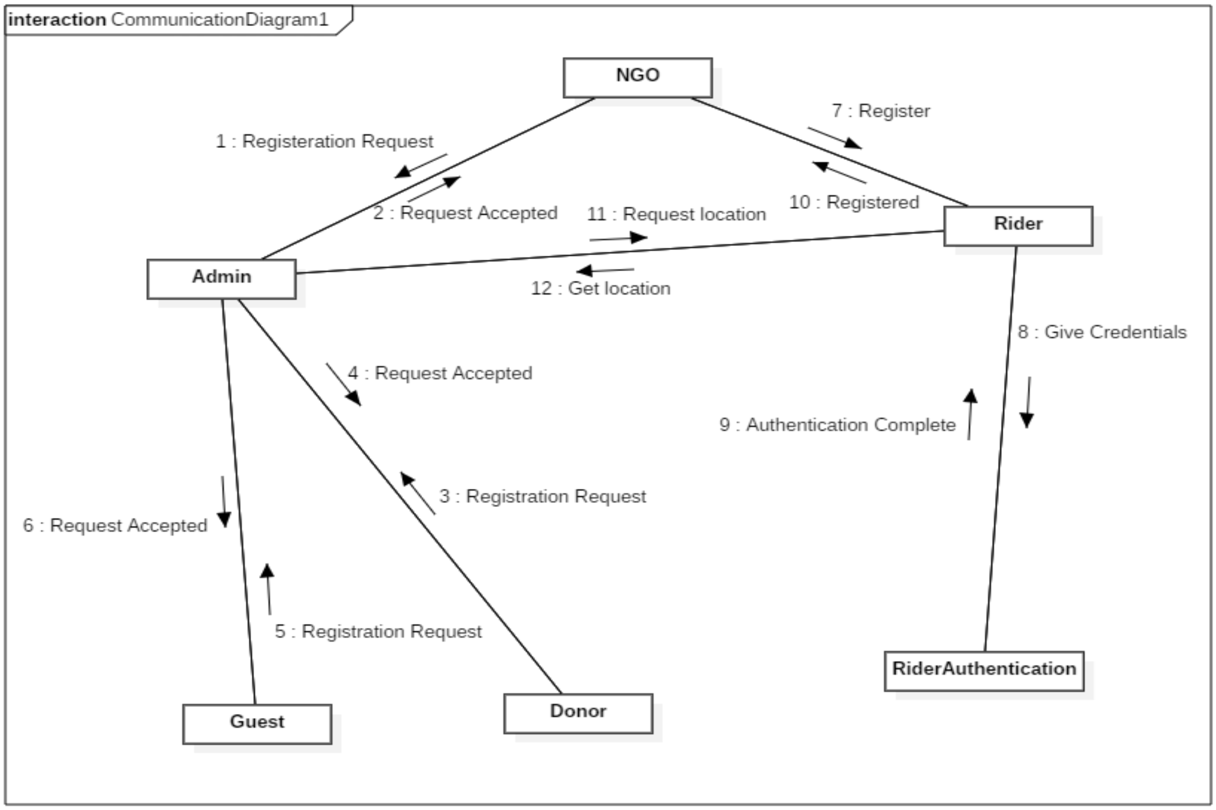


Figure 14: Admin Collaboration Diagram

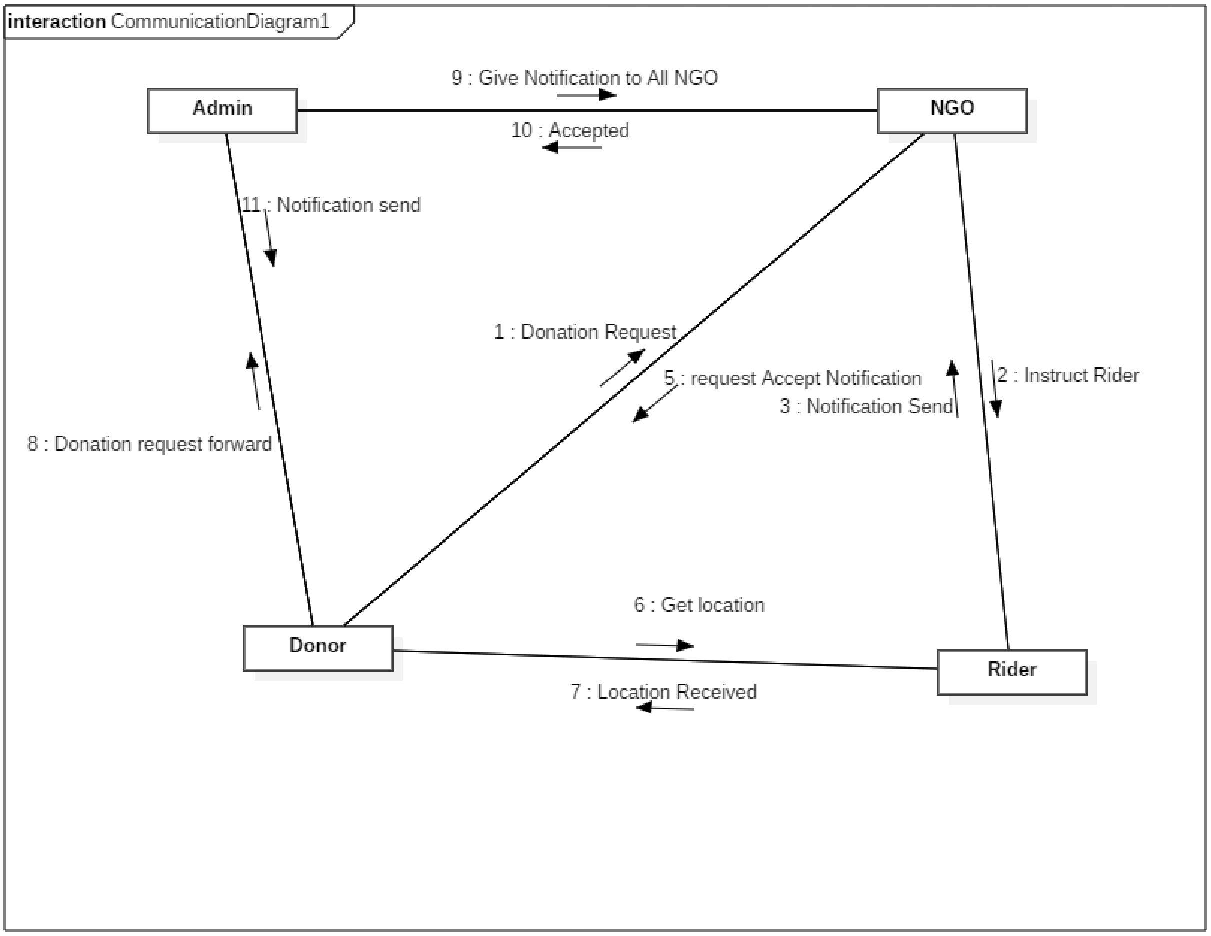


Figure 15: Donor Collaboration Diagram

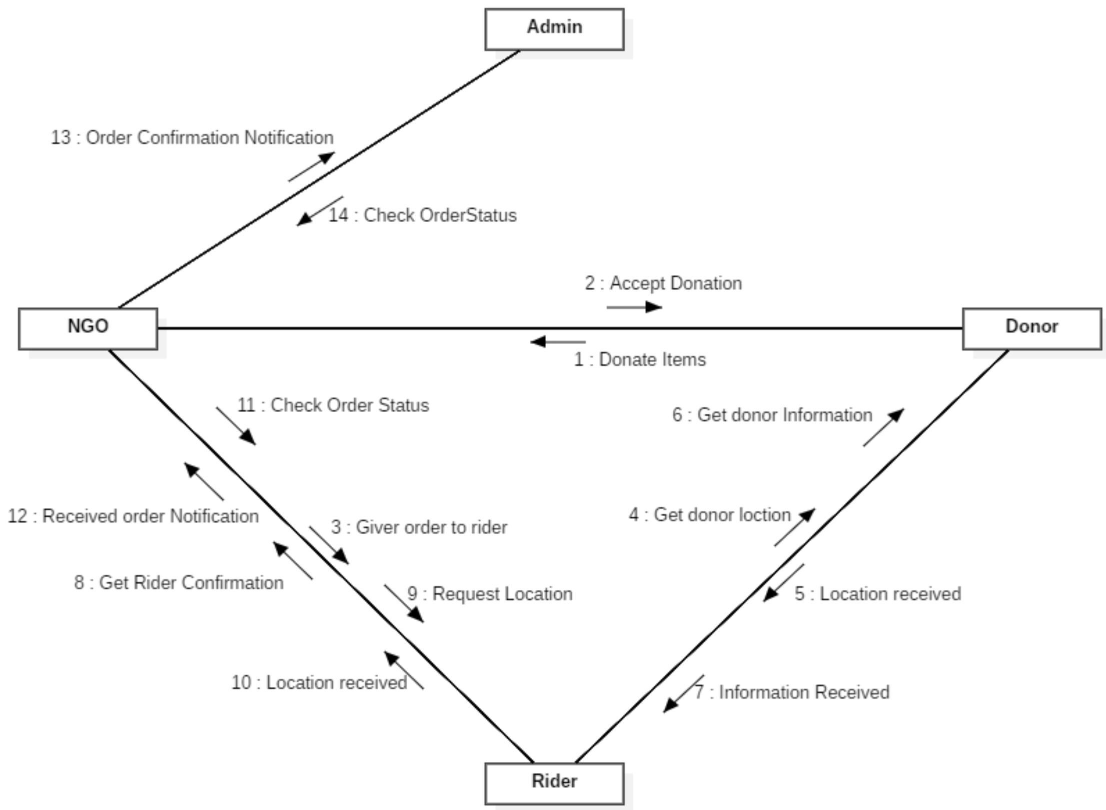


Figure 16: NGO Collaboration Diagram

8.6. Operation Contracts

Use case Description

In the description of use case document contain the following parts:

UC_1:

Brief description	In this use case primary actor “Donor” will register himself in the application and check status
Precondition	Visitor must visit the application.
Basic flow	Donor register himself for donation and check the list of registered NGOs.
Alternate flow	The Admin will not show the unregistered NGOs.
Post conditions	Admin has successfully show the NGOs list.

UC_2:

Brief description	In this use case primary actor “Donor” shall make the login and the secondary actor “Admin” approve the donor login.
Precondition	Admin or Donor have login.
Basic flow	Admin check the donor detail after login of donor
Alternate flow	The Admin will not show the donor list to the admin due to connection failure.
Post conditions	Admin has successfully show the list.

UC_3:

Brief description	In this use case primary actor “Donor” give the donations request and secondary actor “admin” check the request of donor
Precondition	The donor must have a donation request.
Basic flow	The donor generate the donation request and the admin approve this.

UC_4:

Brief description	In this use case primary actor “Donor” after the request the donor check the status of rider for collection of donation.
Precondition	The donor must send the request.
Basic flow	Donor send his request and the admin allocate the rider for collection of donation.

UC_5:

Brief description	In this use case the primary actor “NGO” will login to the Admin and then check the donation detail while the secondary actor “Admin” will save this addition in to the database of the Admin.
Precondition	“NGO” must visit the website.

Basic flow	NGO register himself in the Admin and admin allow to access to the donator
Alternate flow	The Admin will not save the NGO into his database due to any reason.
Post conditions	Admin has successfully add the NGO save into his database.

UC_6:

Brief description	In this use case the primary actor "NGO" will login to the Admin and check the donation detail if it agreed. They send the rider on the donation destination for collection of donation
Precondition	"NGO" must visit the website.
Alternate flow	The Admin will not save the NGO into his database due to any reason
Post conditions	Admin has successfully add the NGO save into his database.

UC_7:

Brief description	In this use case the primary actor "NGO" receive the donation from rider and update the status and "Admin" will update the status in to the database of the Admin.
Precondition	"Rider" must visit the App.
Alternate flow	The Admin will not save the Donation detail into his database due to any reason.
Post conditions	Admin has successfully add the Donation detail save into his database.

UC_8

Brief description	In this use case the primary actor "Rider" have login in the app and "Admin" approve his login in the Admin.
Precondition	"rider" registration is pending from Admin
Alternate flow	The Admin will not save the rider into his database due to any reason.
Post conditions	Admin has successfully add the rider save into his database.

UC_9

Brief description	In this use case the primary actor "rider" get the order from NGO and update the status and "Admin" will update the status in to the database of the Admin.
Precondition	"Rider" must visit the App.
Alternate flow	The Admin will not save the rider detail into his database due to any reason.
Post conditions	Admin has successfully add the rider detail save into his database.

UC_10

Brief description	In this use case the primary actor "Rider" receive the donation from Donor and update the status and "Admin" will update the status in to the database of the Admin.

Precondition	“Rider” must visit the App.
Alternate flow	The Admin will not save the Donation into his database due to any reason.
Post conditions	Admin has successfully add the Donation save into his database.

UC_11

Brief description	In this use case the primary actor “rider” deliver the donation in NGO and update the status and “Admin” will update the status in to the database of the Admin.
Precondition	“Rider” must visit the App.
Alternate flow	The Admin will not save the Donation detail into his database due to any reason.
Post conditions	Admin has successfully add the Donation detail save into his database.

UC_12

Brief description	In this use case the primary actor “GUEST” makes a login and “Admin” updates this
Precondition	“Guest” must visit the App.
Alternate flow	The Admin will not save the Guest detail into his database due to any reason.
Post conditions	Admin has successfully add the Guest detail save into his database.

UC_13

Brief description	In this use case the primary actor “Guest” give the location
Precondition	“Guest” must visit the App.
Alternate flow	The Admin will not save the Guest into his database due to any reason.
Post conditions	Admin has successfully add the Guest save into his database.

UC_14

Brief description	In this use case the primary actor “Guest” get the donation
Precondition	“Guest” wait for donation.
Alternate flow	The NGO will send the donation to the guest.
Post conditions	NGO has successfully send the donation

8.7. Design Class Diagram

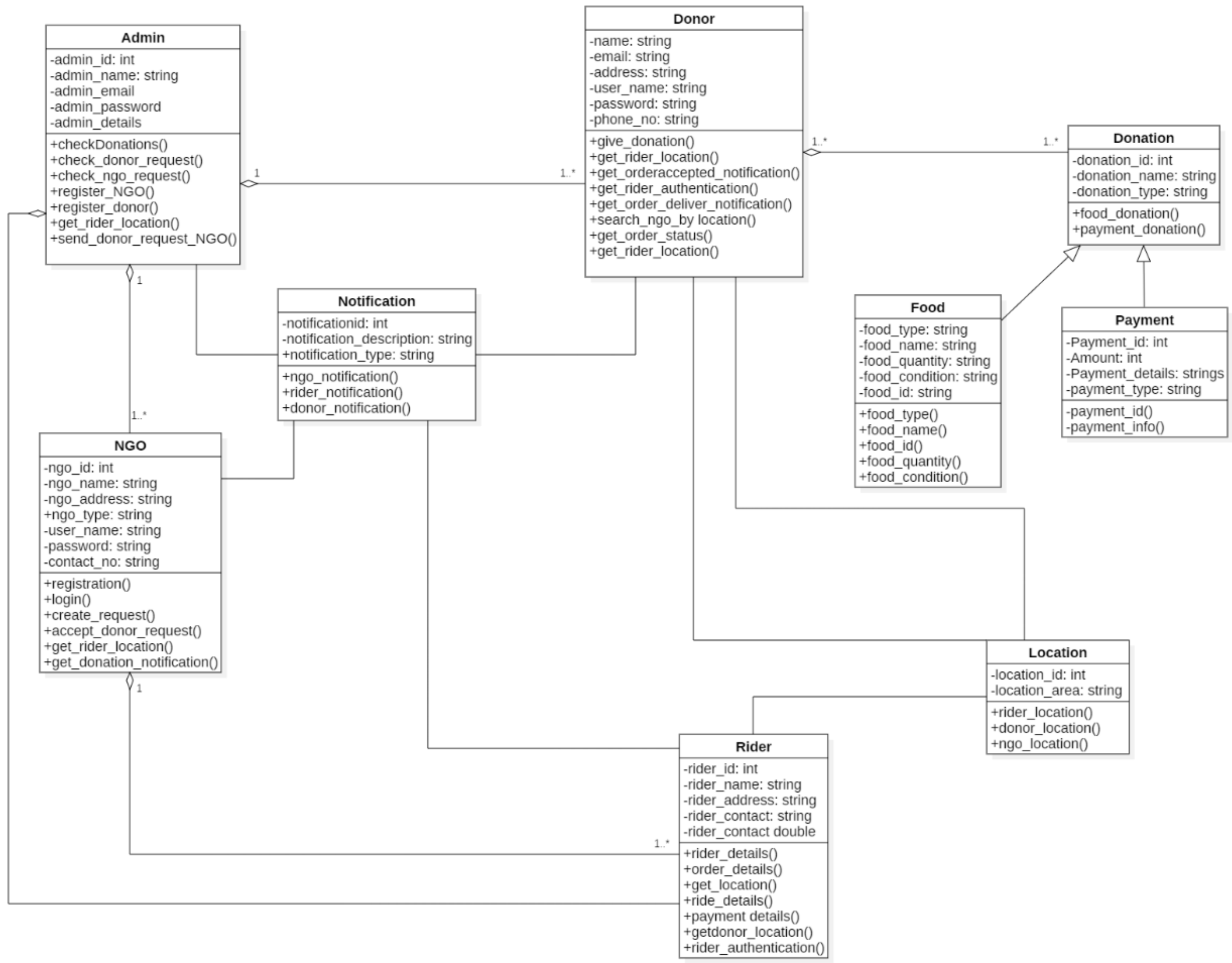


Figure 17: Class Diagram

8.8. State chart diagram

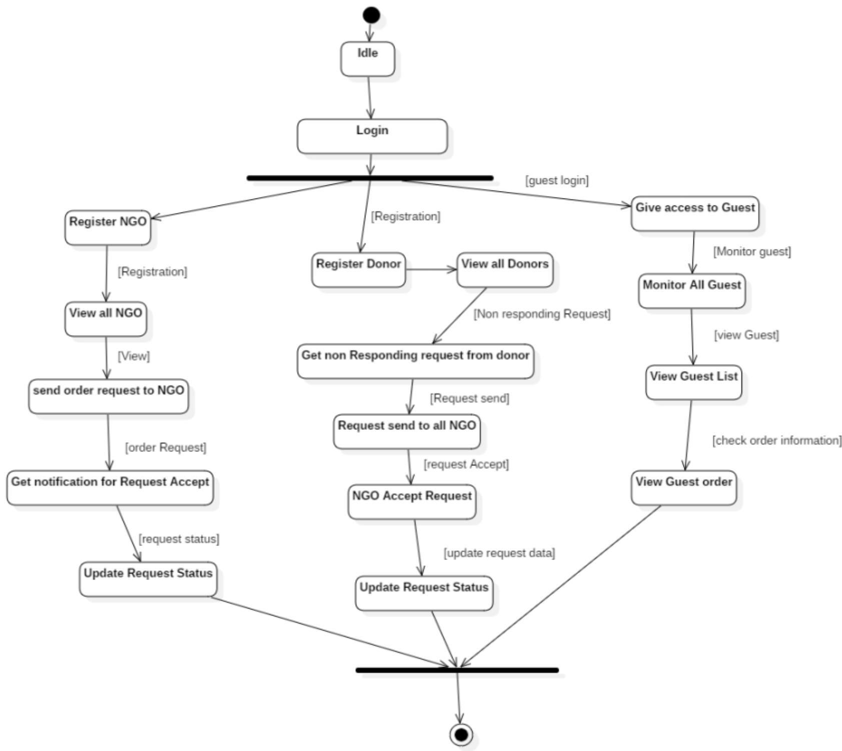


Figure 18: Admin State chart Diagram

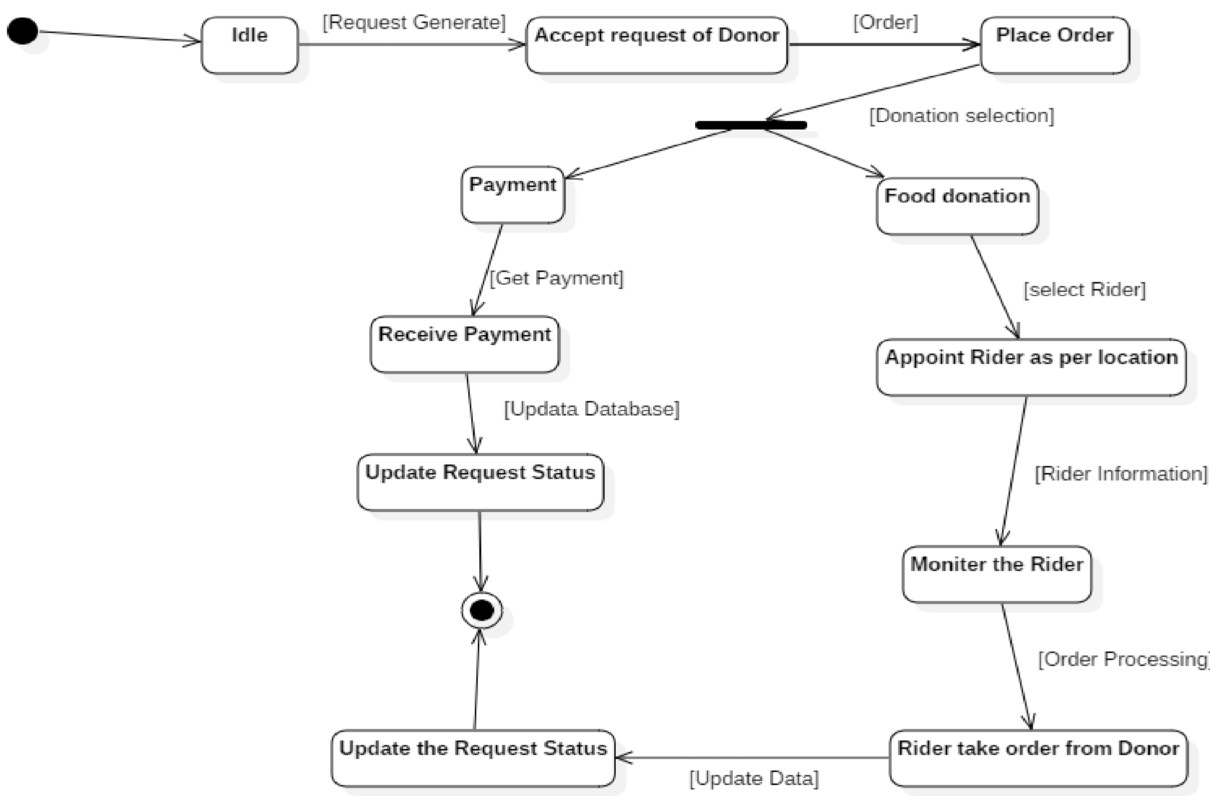


Figure 19: NGO state chart Diagram

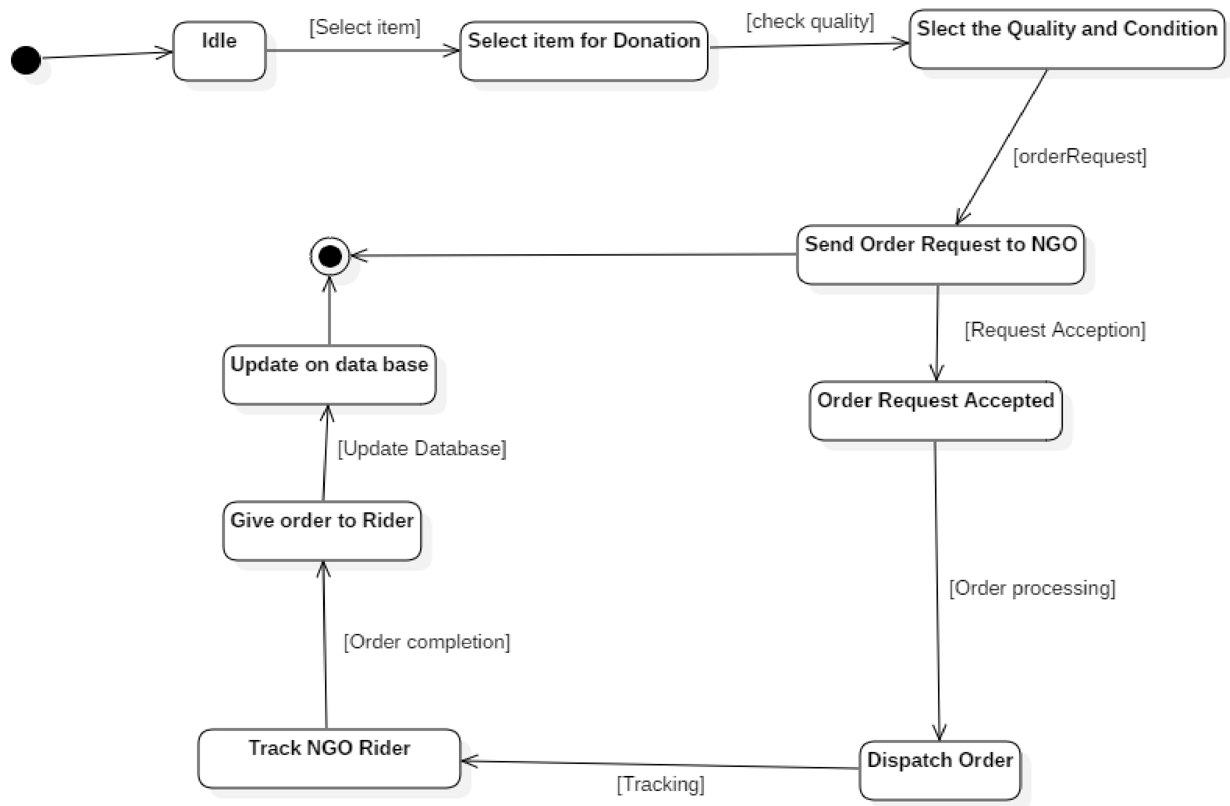


Figure 20: Donor State Chart Diagram

8.9. Data Model

The data model is a subset of the implementation model, which describes the logical and physical representation of persistent data in the system.

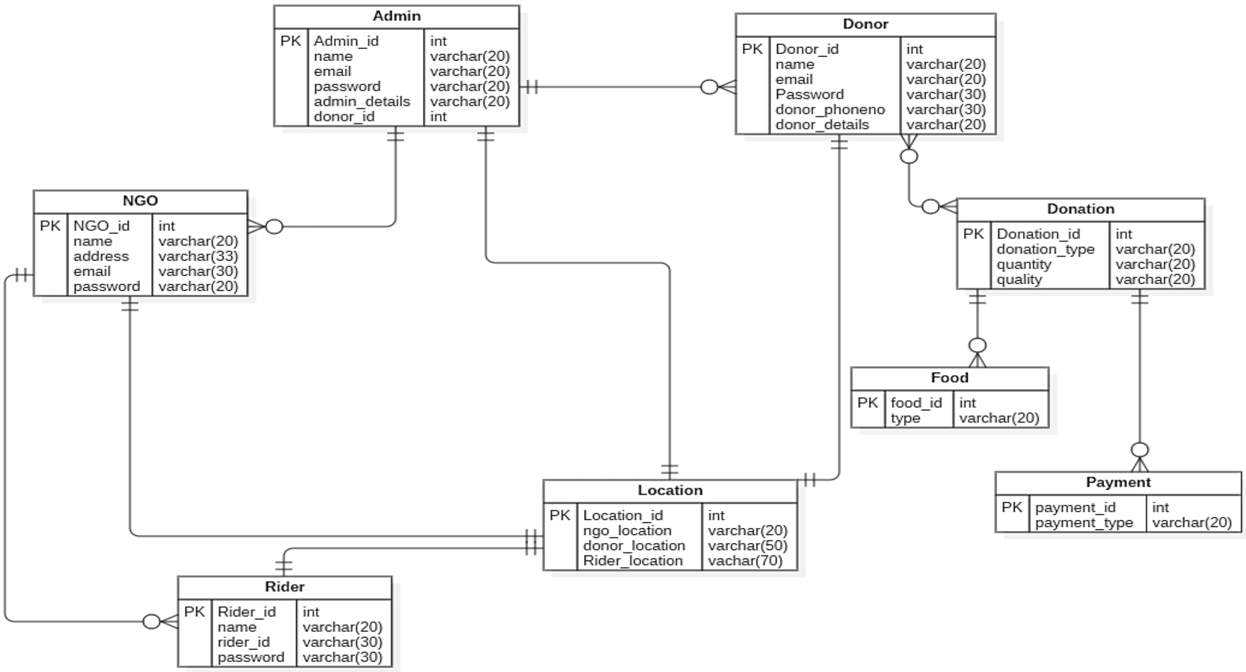


Figure 21: ERD Diagram

Chapter No 9: Requirements Traceability Matrix

9.1 Requirements Traceability Matrix

Table 12: Requirements Traceability matrix

Sr.#	Para#	Initial Requirements	Build	Use Case Name	Category
1	1.0	User shall register himself asDonor	B1	UC_Registration Request	Donation
2	1.0	System Admin "shall" view registration.	B1	UC_View_ registration	Donation
3	1.0	System "shall" accept or reject therequest	B1	UC_Accept/Reject User_Request	Donation
4	1.0	A User "shall" login to the system	B1	UC_Login	Donation
6	1.0	User "shall" search for NGOs	B1	UC_Search	Donation
7	1.0	A User "will" place Request for Donation	B1	UC_Place_Request	Donation
8	1.0	A User "will" edit Request forride	B1	UC_Edit_	Donation

Chapter No 10: Risk Analysis

10.1 Risk List

- Time Constraint
- Unavailability of resources
- Lack of expertise
- Ignorance of non-functional requirements
- Risk due to complexity of the project
- Technology Revolution
- Requirements are not clear. The system is difficult to develop because the requirements are changed by the time.
- Gujrat is an underdeveloped city and the people in this city are mostly computer illiterate.
- Due to bribery of readers it is very difficult to implement or deploy the system.
- The system requires updating.
- It is difficult to upload whole previous record.
- Our system does not cause any harm to peoples' lives, property etc. Our system is intended to be deployed for donation purpose only. It is, under no circumstances, advised to completely depend on this information.
- Moreover, there is a risk of servers being busy due to too many user requests. There is always a risk involved in digital systems. There can be multiple bugs and security attacks on databases.

Chapter No 11: Cost Estimation Sheet

11.1 Project/Product Costing

11.1.1. Project Cost Estimation By Function Point Analysis

In Project Cost Estimation by Functional Point Analysis, following five information domain characteristics are determined and counts are provided in here as:

Count of total external inputs = 23
Count of total external output = 23
Count of total user inquiries = 8
Count of total internal logical files = 9
Count of total external interface files = 6

Information Domain Values	Low Values	Average Values	High Values	FP Count
No. of External Inputs	3 x 3 = 9	12 x 4 = 48	8 x 6 = 48	105
No. of External Outputs	3 x 4 = 12	12 x 5 = 60	8 x 7 = 56	128
No. of User Inquiries	0 x 3 = 0	4 x 4 = 16	4 x 6 = 24	40
No. of Internal Logical Files	3 x 7 = 21	2 x 10 = 20	4 x 15 = 60	101
No. of External Interface Files	3 x 5 = 15	1 x 7 = 7	2 x 10 = 20	42
Total				416

Factors	Values (0 - 5)
Data Communications	5
Distributed Data Processing	3
Performance	5
Heavily Used Configuration	2
Transition Rate	4

Online Data Entry	2
End-user Efficiency	4
Online Update	1
Complex Processing	0
Reusability	2
Installation Ease	3
Operational Ease	5
Multiple Sites	1
Facilitate Change	3

Total = Σ F_i = 40

FP est. = Count Total * [0.65 + 0.01 * (Fi)]
FP est. = 416 * [0.65 + 0.01 * (40)]
FP est. = 416 * [0.65 + 0.4]
FP est. = 416 * [1.05]
FP est. = 436.8
FP est. = 437

Finally, Total Project Cost and Total Project Effort are calculated given the average productivity parameter for the system.

The formulas are given as follows:

For our project
Average Productivity=34 fpm
Labor Rate=12000 per month
Total Estimated Effort=FP est./Productivity
Total Estimated Effort =437/34
Total Estimated Effort=12.85 pm
Now,
Cost/FP=Labor Rate/Productivity
Cost/FP=12000/34
Cost/FP=352.94 Rs/FP
Total project cost=FP est.*(Cost/FP)
Total project cost=437*(352.94)
Total project cost=154235.294pkr
Total project cost=160000 (approx.)

11.1.2 Project Cost Estimation by using COCOMO’81 (Constructive Cost Model)

Basic COCOMO

Type	Effort	Schedule
Organic	PM= 2.4 (KLOC)1.05	TD= 2.5(PM)0.38
Semi-Detached	PM= 3.0 (KLOC)1.12	TD= 2.5(PM)0.35
Embedded	PM= 2.4 (KLOC)1.20	TD= 2.5(PM)0.32

PM= person-month (effort)

KLOC= lines of code, in thousands

TD= number of months estimated for software development (duration)

Basic COCOMO

Type : Semi-Detached

KLOC = 10,000L=10K;

Salary per month of each developer = 10,000

Effort = a1 x (KLOC) a2PM

Effort : 3.0 (10) ^1.12=39.54(man/month)

T. Dev = b1 x (Effort) b2 Months

T. Dev= 2.5(39.54)0.35=9.06(months)

People Required = 39.54/ 9.06=4

Total cost = people required *salary

Total cost = 4*10,000

Total cost = 40,000

11.1.3. Activity Based Costing

Activity	Resources	Cost Rate	Duration (Weeks)
A) Requirement gathering, Analysis and Design Diagrams	Star UML	2.5k	4
B) Layout Design	Android Studio	1.5k	1
C) Design System and User panels	Android studio	1.5k	2
D) Map Integration	Google Maps	2k	1

E) Database Connectivity and APIs for application	Android Studio and Firebase	2k	2
F) Integrate all modules		1.5k	3
G) Testing		1k	3

Chapter No 12: References

https://developer.android.com/studio/

https://app.diagrams.net/

https://www.flaticon.com/

https://icons8.com/